

Phylogenetics

Introduction to molecular
dating

RL-V3 MPP

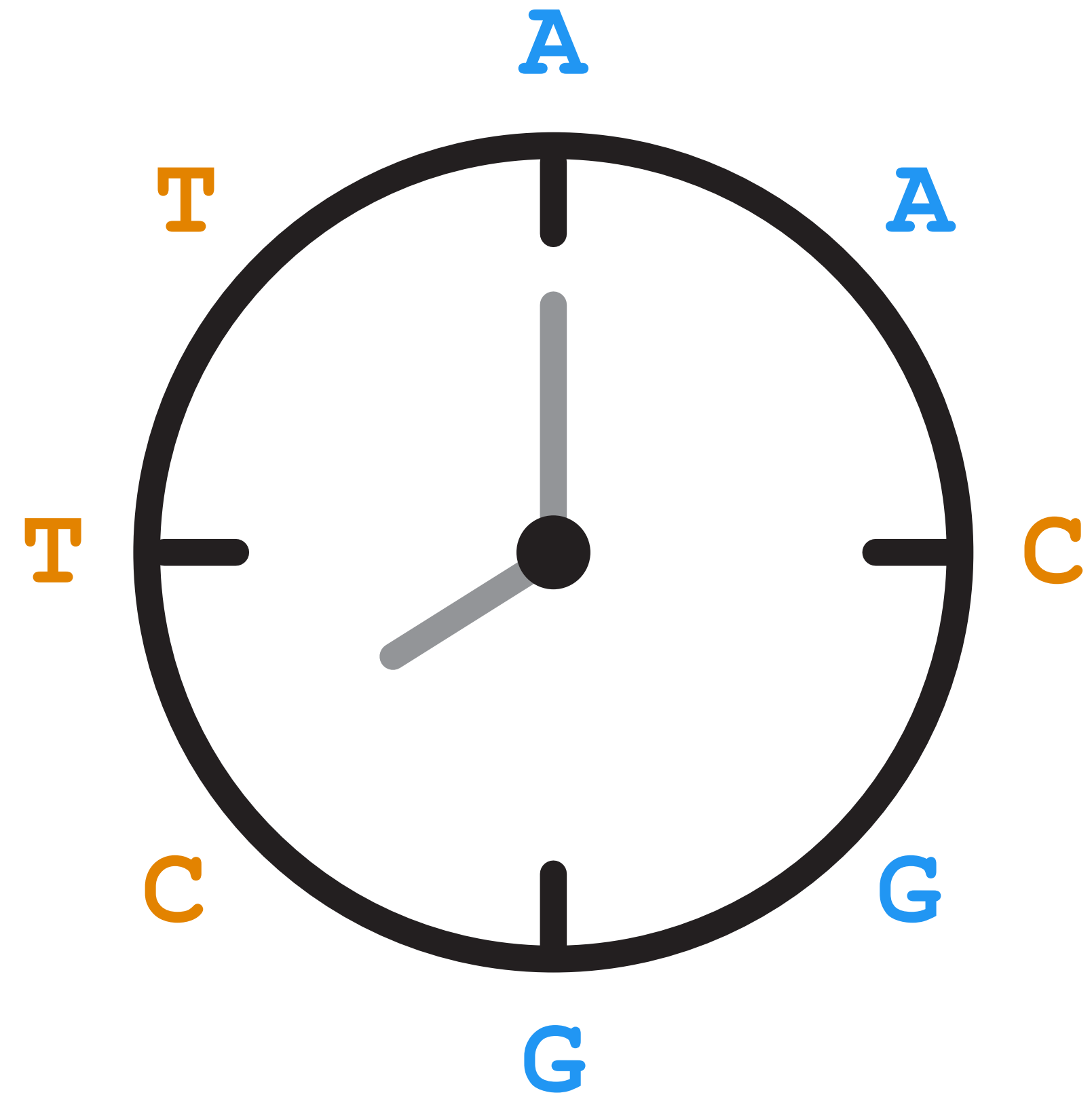
Rachel Warnock

16.05.25



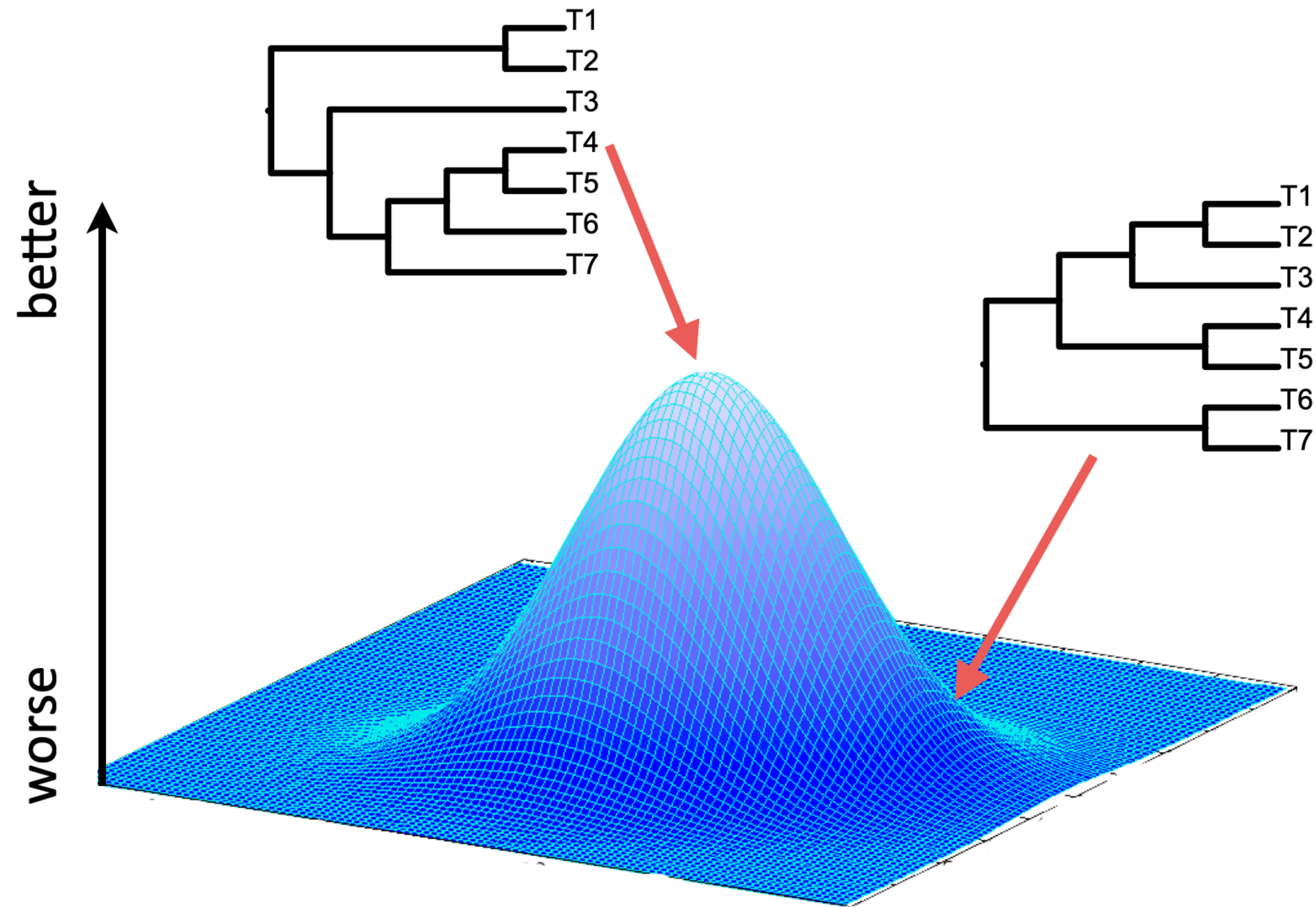
Today's objectives

- Molecular dating
- Intro to BEAST2



Recap

How do we find the 'best' tree?



It depends how you measure 'best'

Method	Criterion (tree score)
Maximum parsimony	Minimum number of changes
Maximum likelihood	Likelihood score (probability), optimised over branch lengths and model parameters
Bayesian inference	Posterior probability, integrating over branch lengths and model parameters

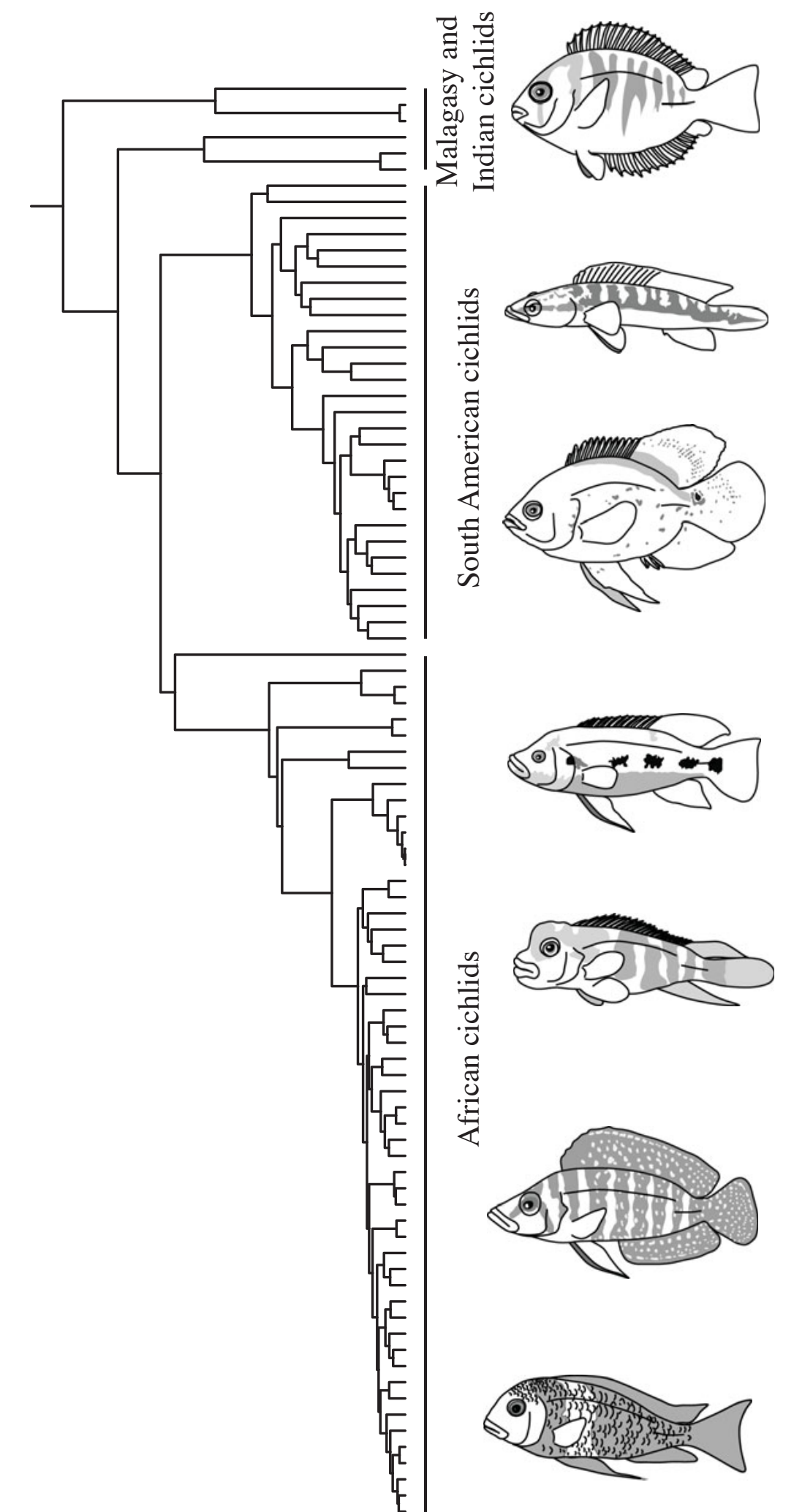
Both maximum likelihood and Bayesian inference are model-based approaches

Note these are not the only approaches to tree-building but they are the most widely used

Introduction to molecular dating

What can we learn from trees?

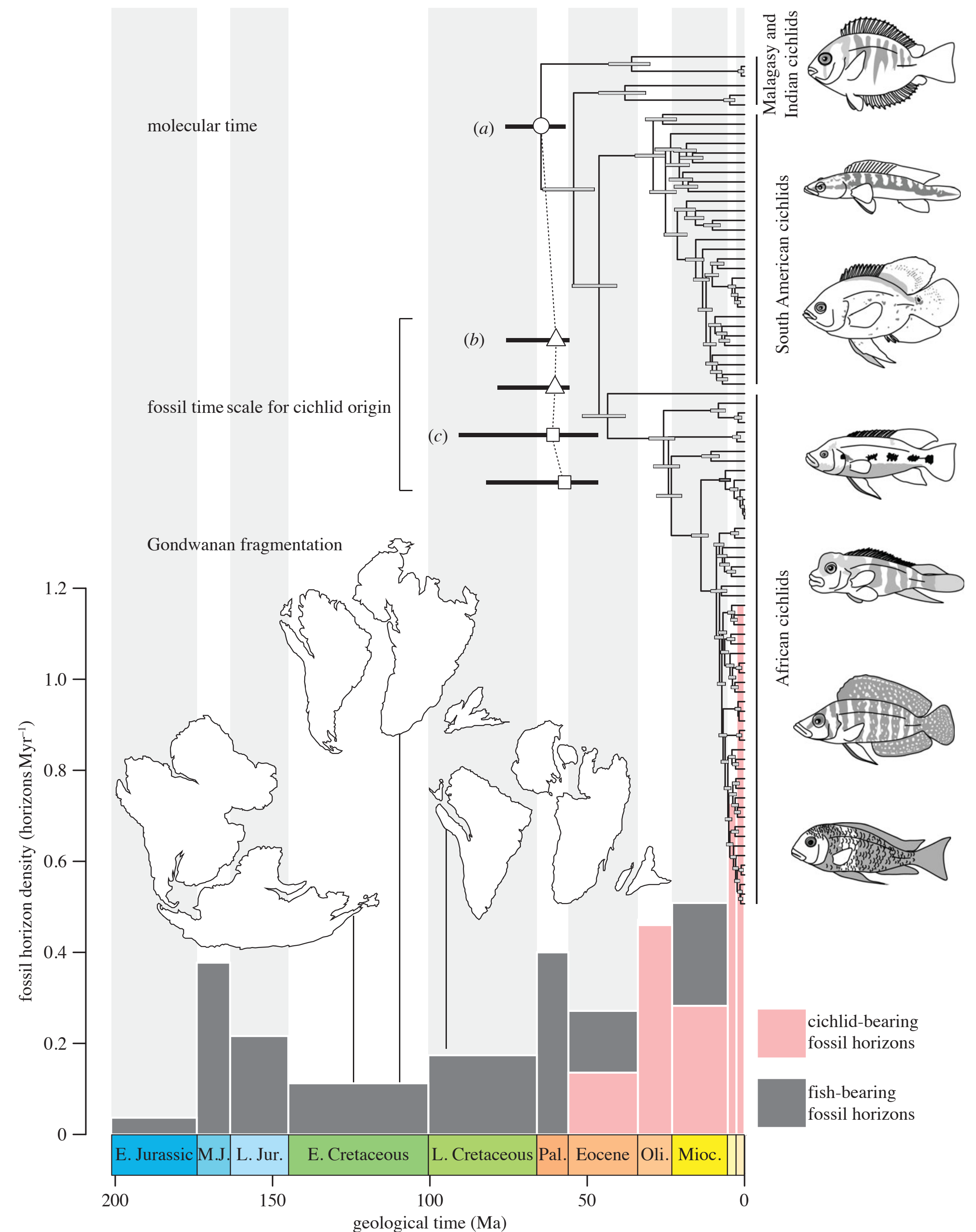
- Evolutionary relationships



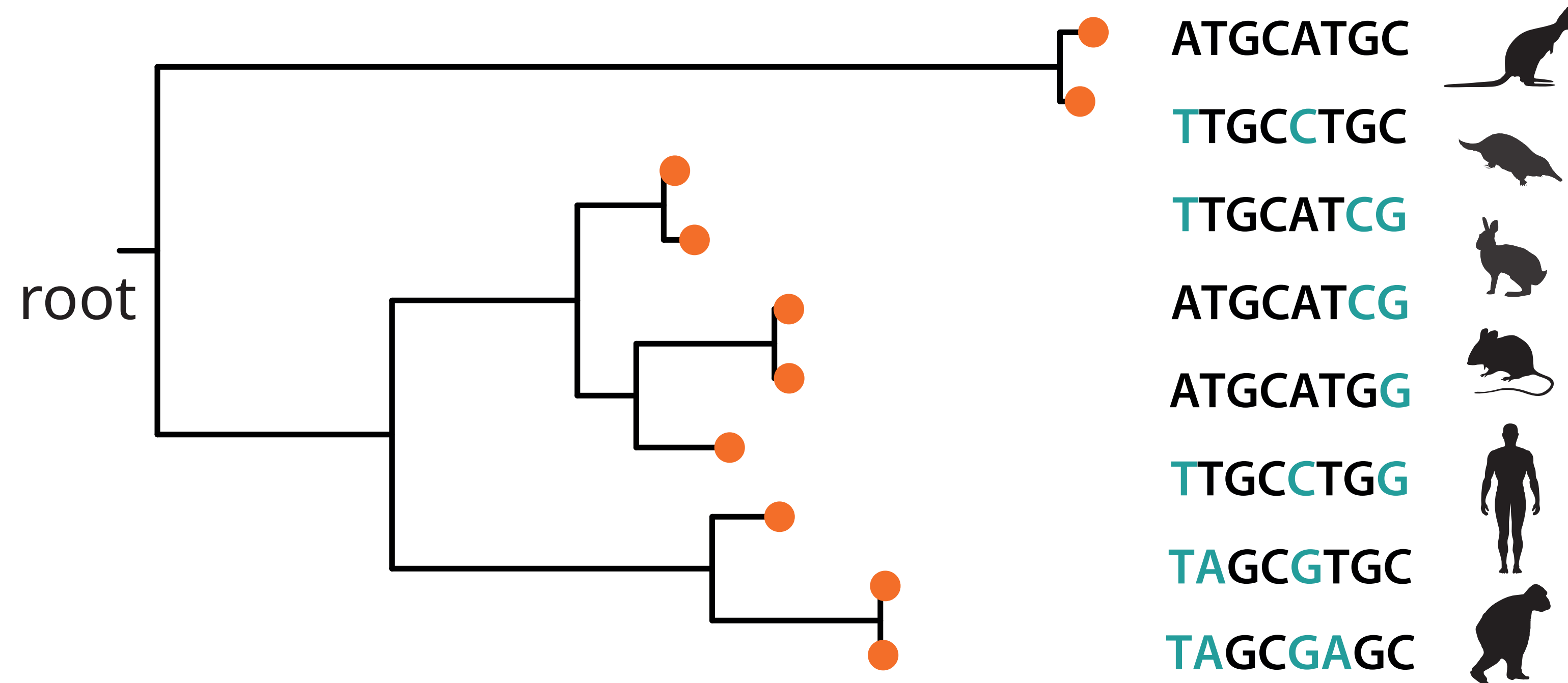
What can we learn from trees?

- Evolutionary relationships
- Timing of diversification events
- Geological context
- Rates of phenotypic evolution
- Diversification rates

Image adapted from *Friedmann et al.* ([2013](#))



Molecular (or morphological) characters are not independently informative about time



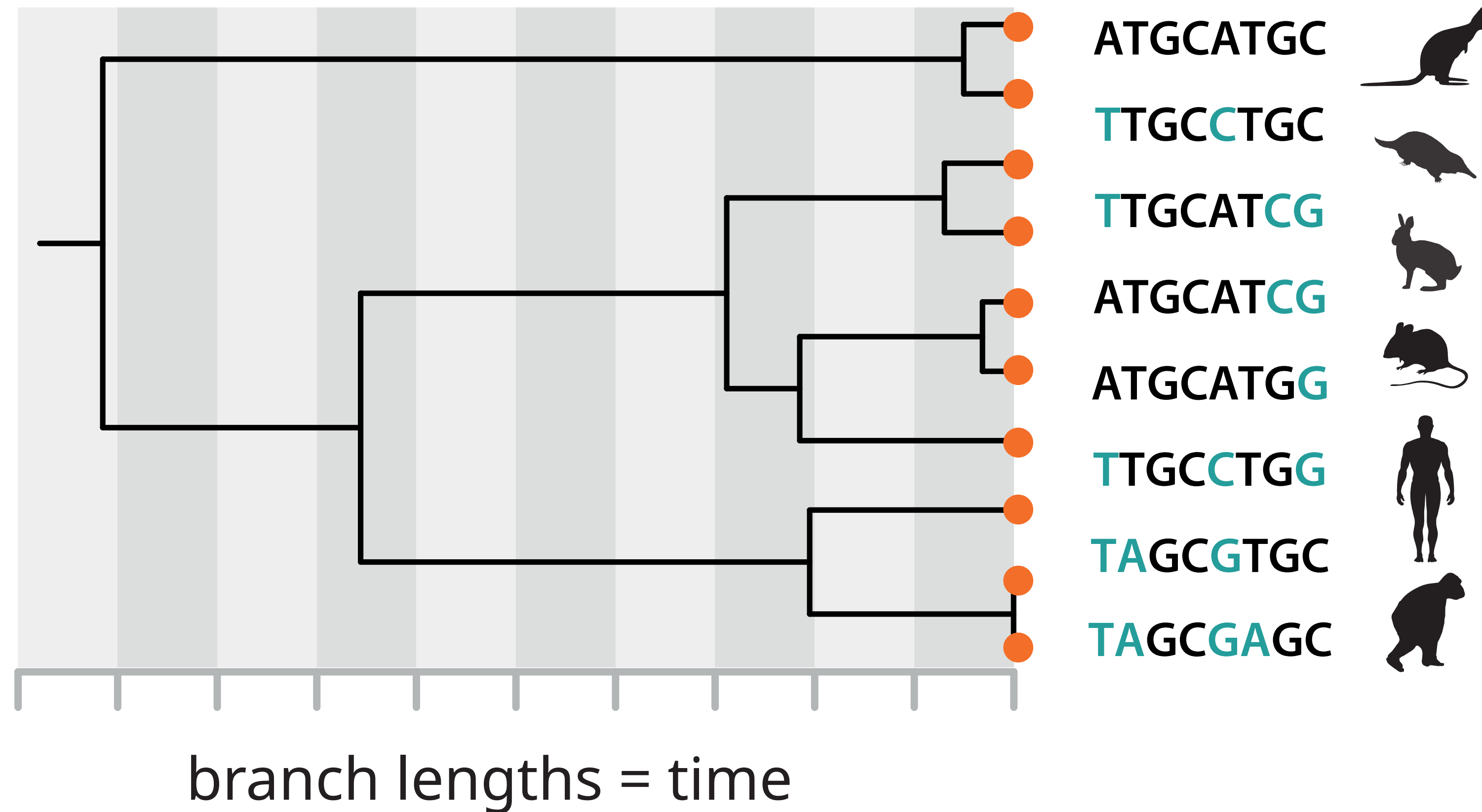
Slow rate, long interval **or** fast rate, short interval?

Goal: to disentangle evolutionary rate and time

branch lengths = genetic distance

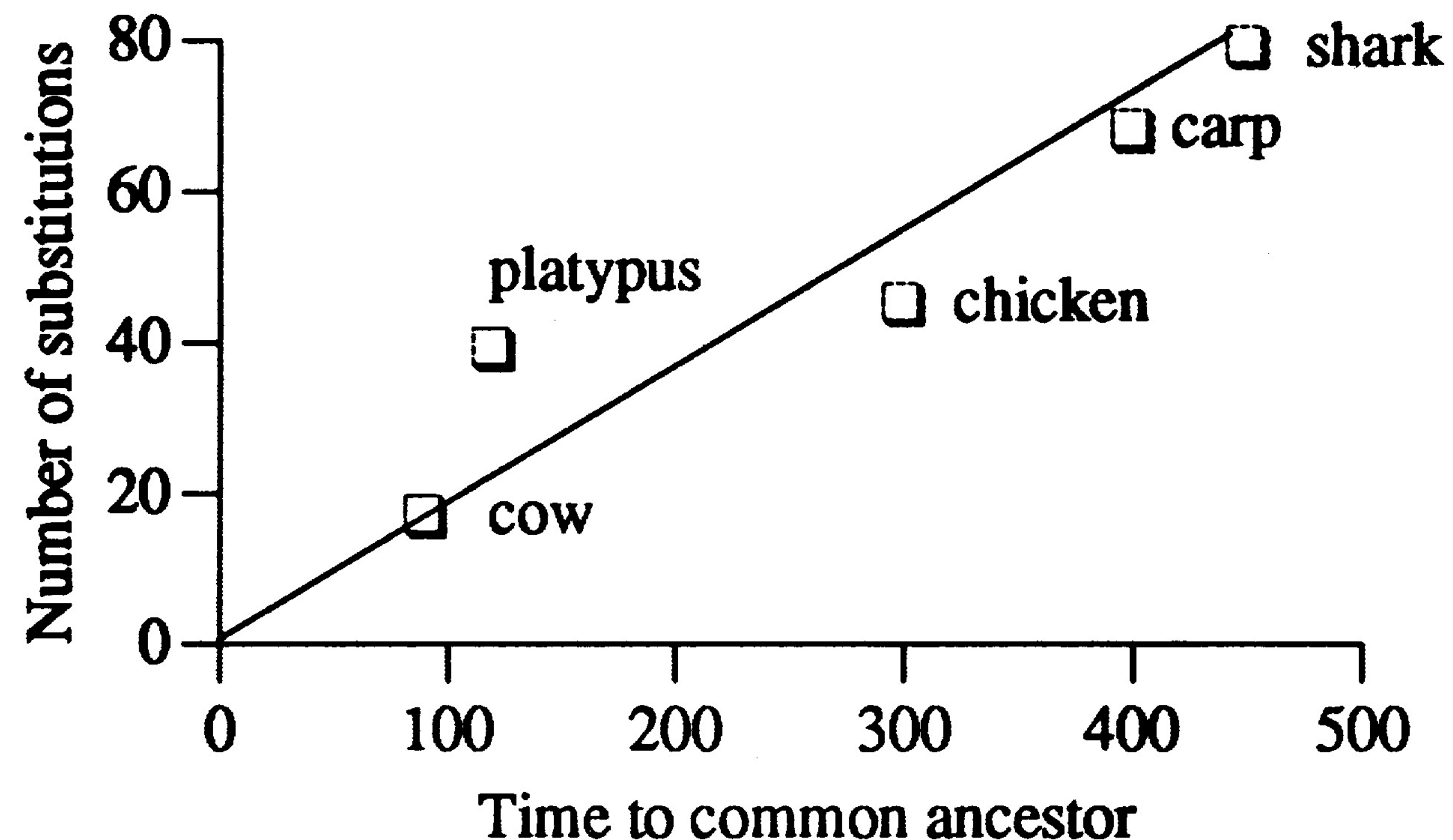
$$v = rt$$

Molecular (or morphological) characters are not independently informative about time



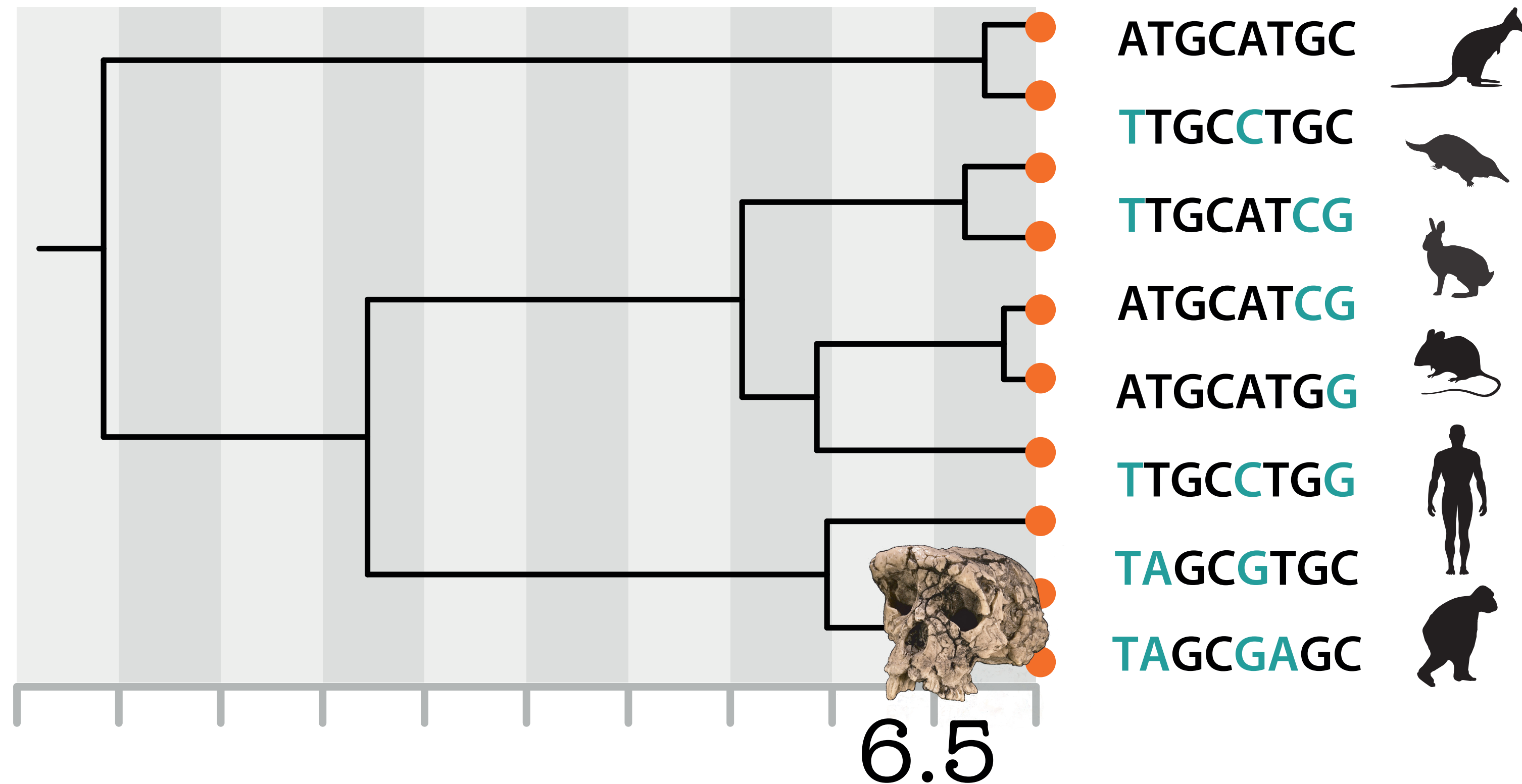
Goal: to disentangle evolutionary rate and time

The molecular clock hypothesis



Molecules as documents of evolutionary history *Zuckerkandl & Pauling* ([1965](#))
A history of the molecular clock *Morgan* ([1998](#))

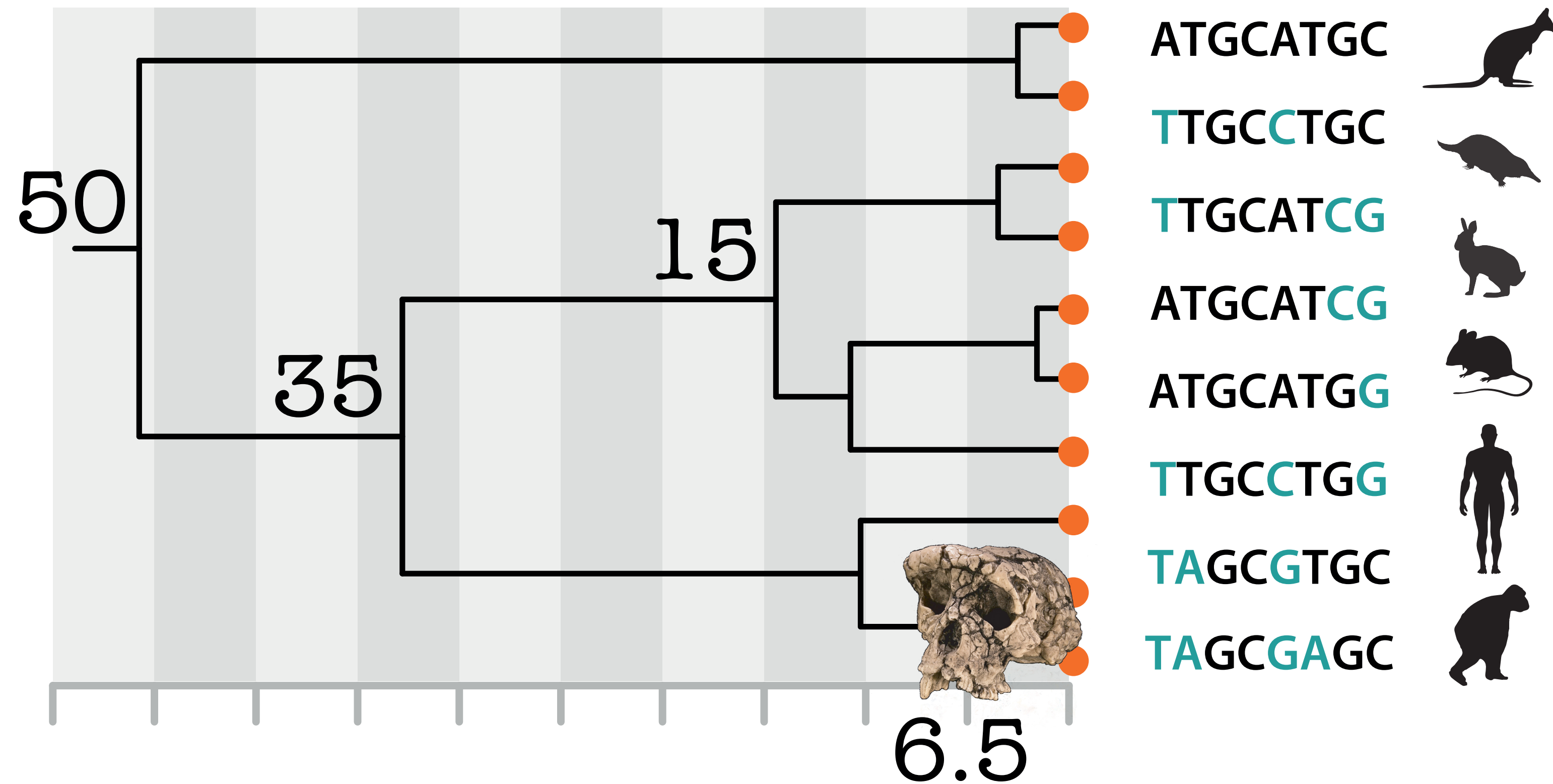
Calibrating the substitution rate



branch lengths = time

Temporal evidence of divergence for one species pair let's us **calibrate** the average rate of molecular evolution

Calibrating the substitution rate

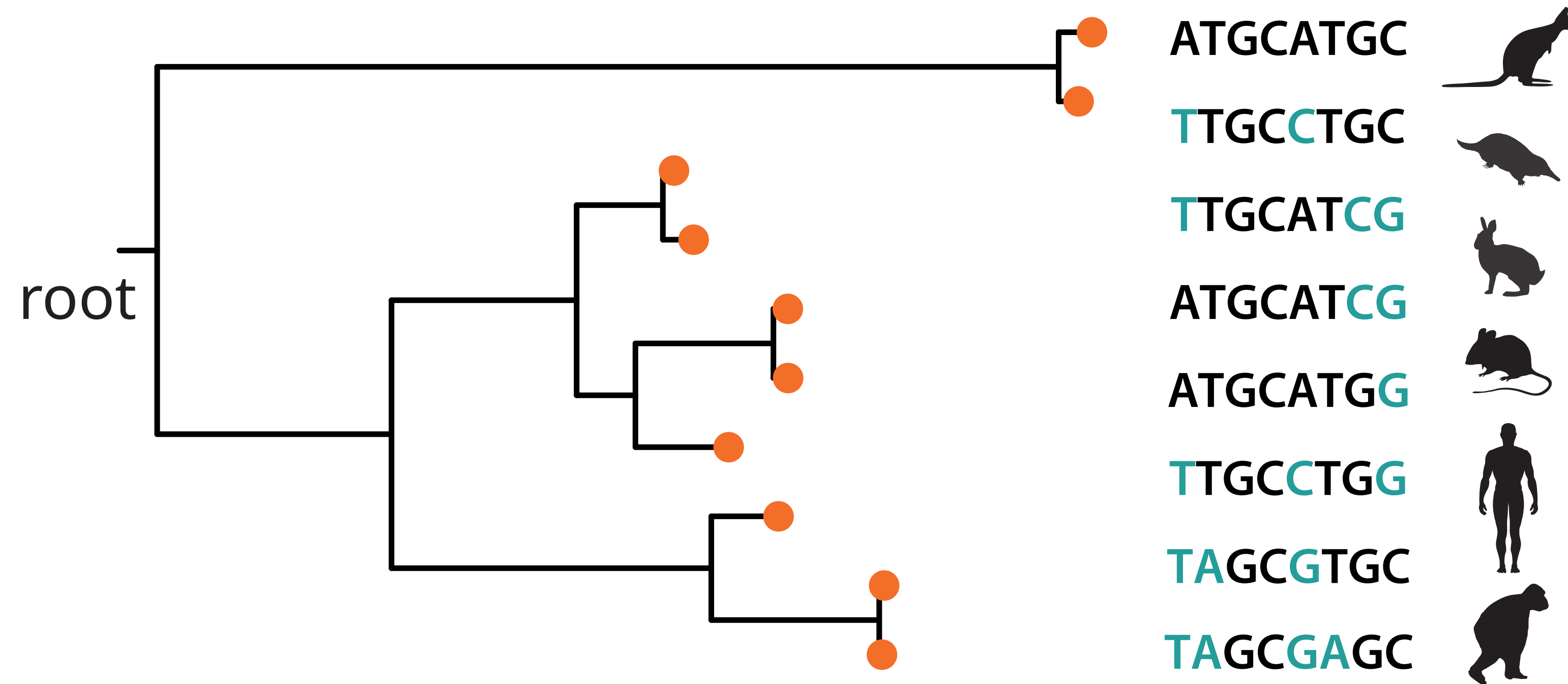


branch lengths = time

We can use this rate to extrapolate the divergence times for other species pairs

Molecular dating: challenges

Rate and time are not **fully identifiable**!

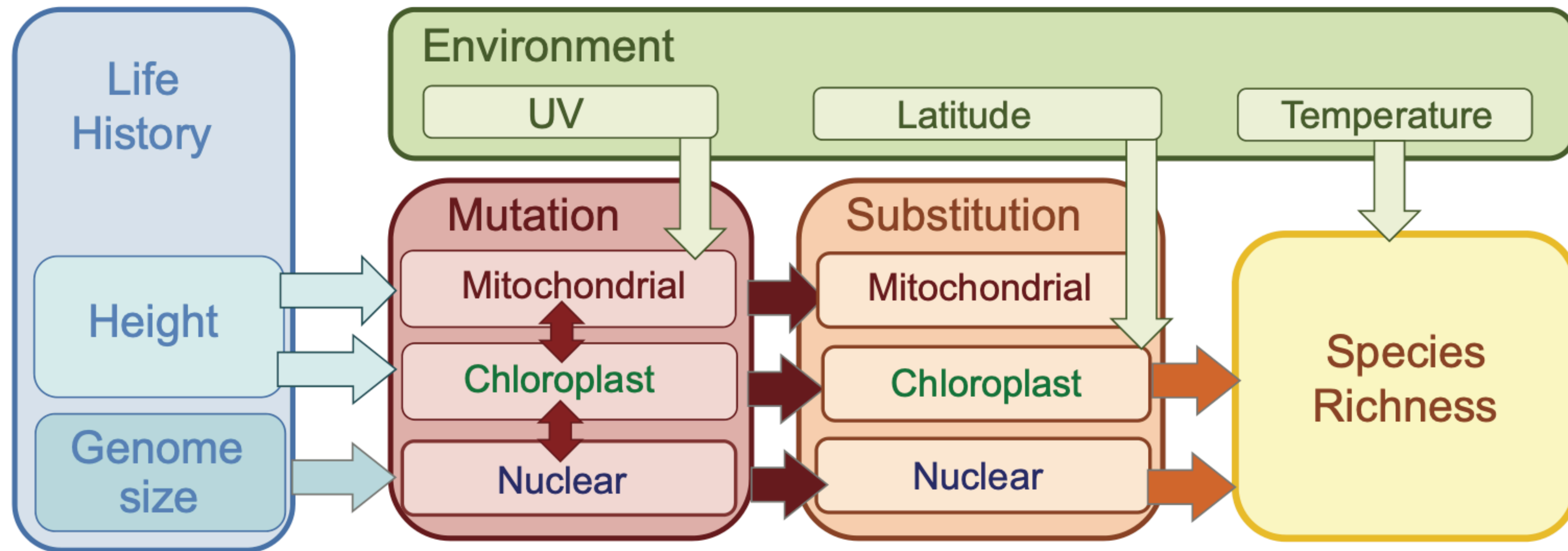


branch lengths = genetic distance

$$v = rt$$

Molecular dating: challenges

Many variables contribute to **variation in the substitution rate**



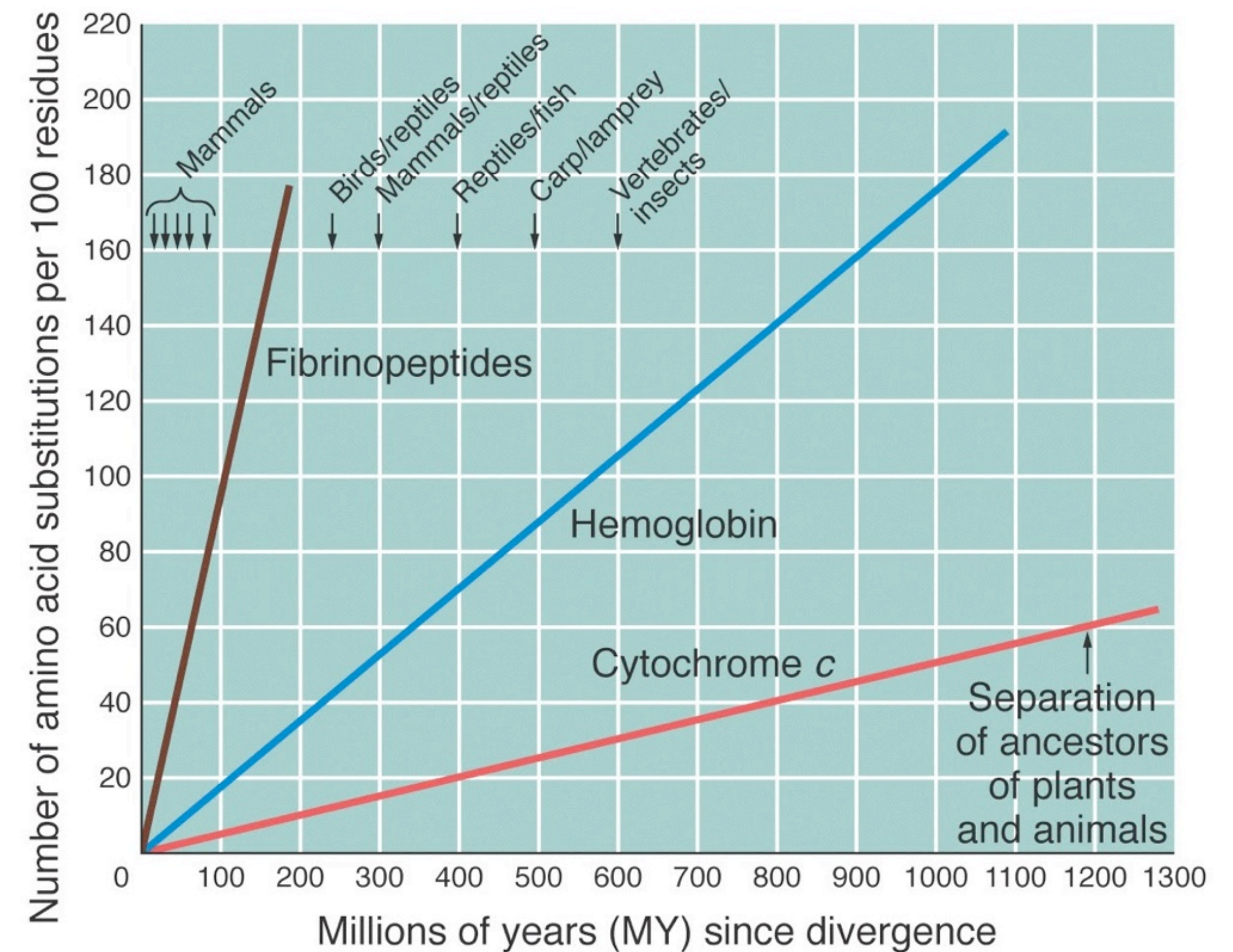
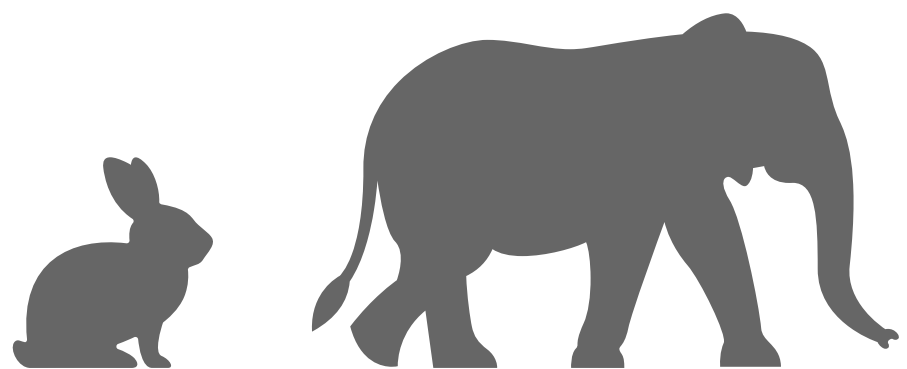
Molecular dating: challenges

Many variables contribute to **variation in the substitution rate**

The molecular clock is not constant

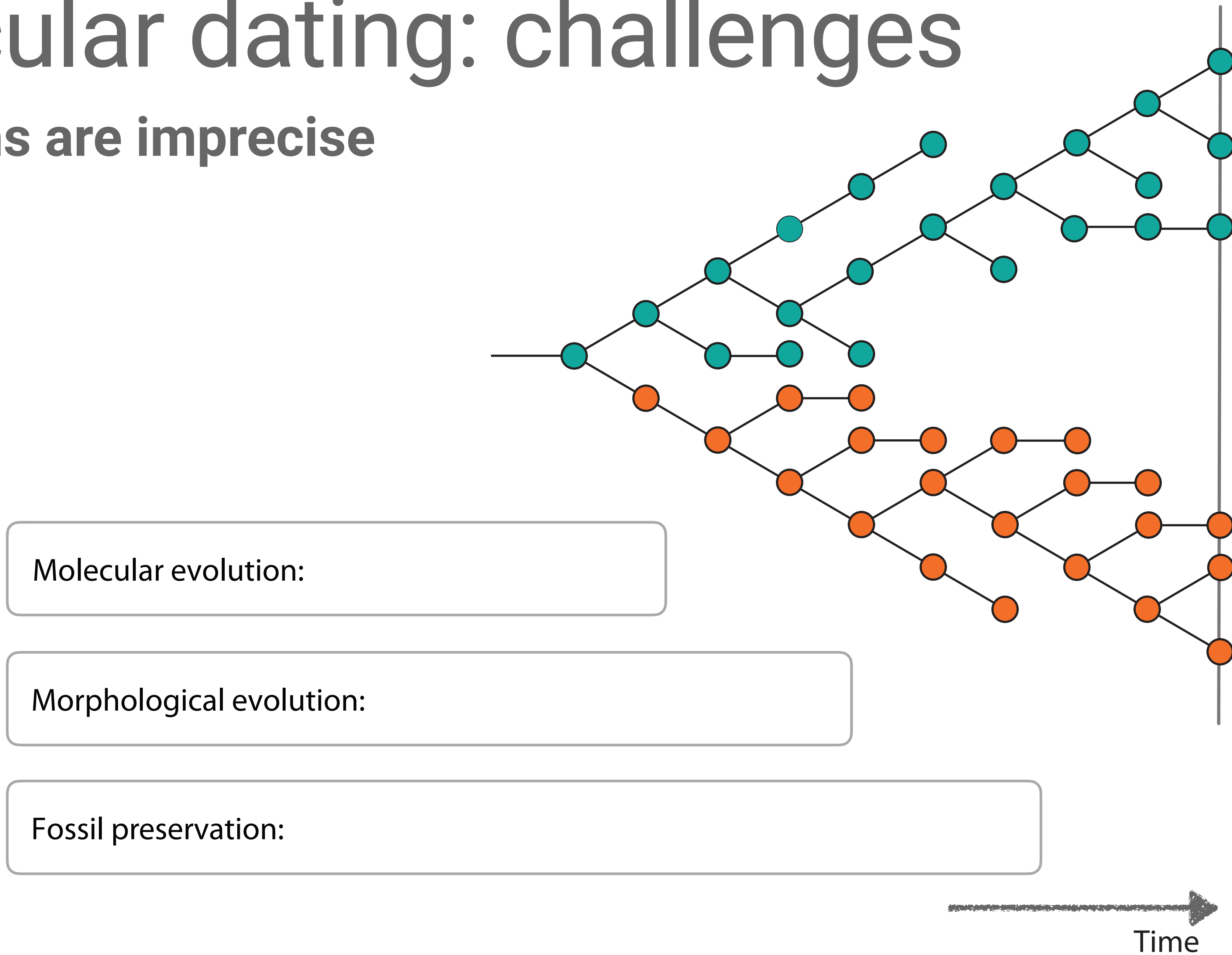
Rates vary across:

- taxa
- time
- genes
- sites within the same gene



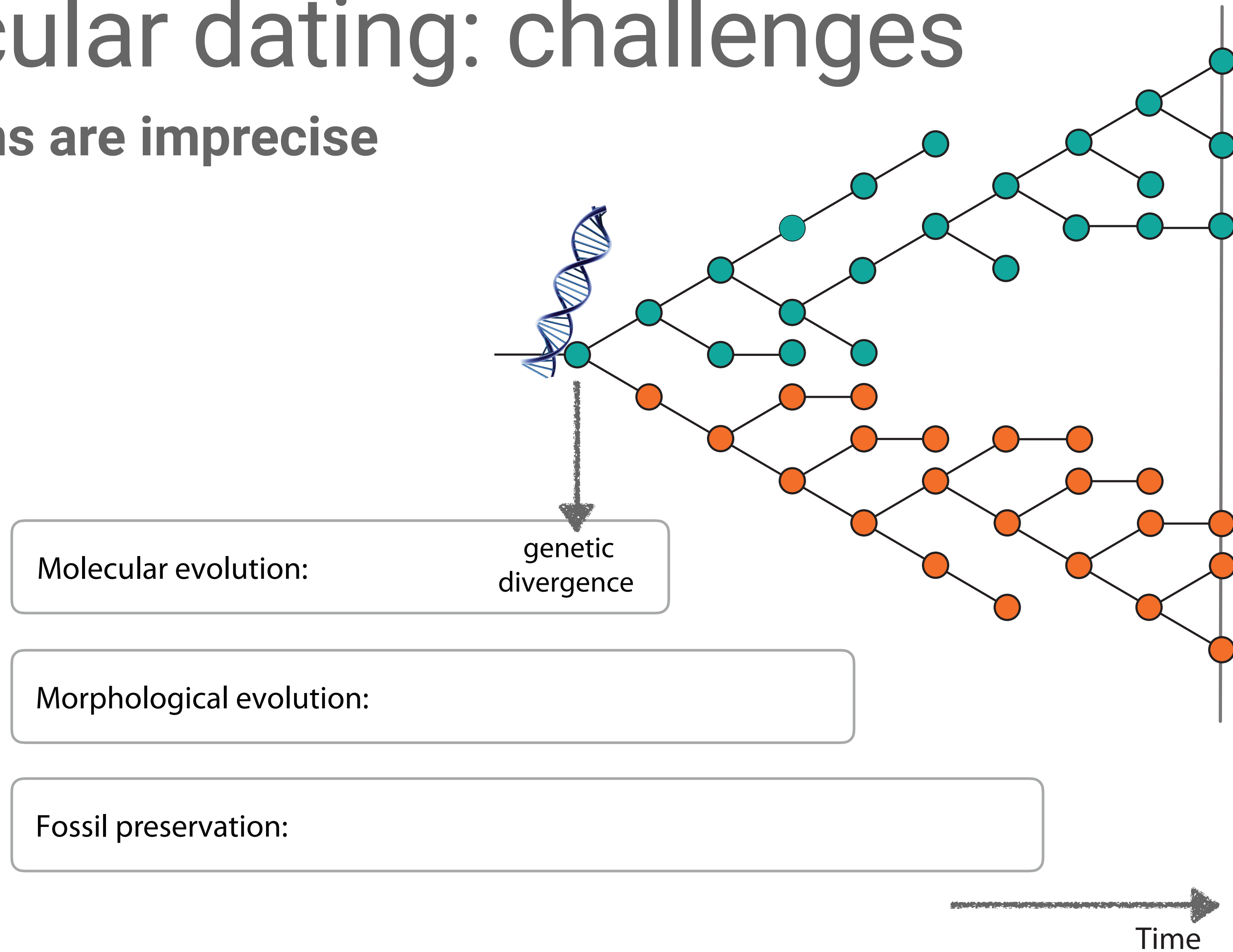
Molecular dating: challenges

Calibrations are imprecise



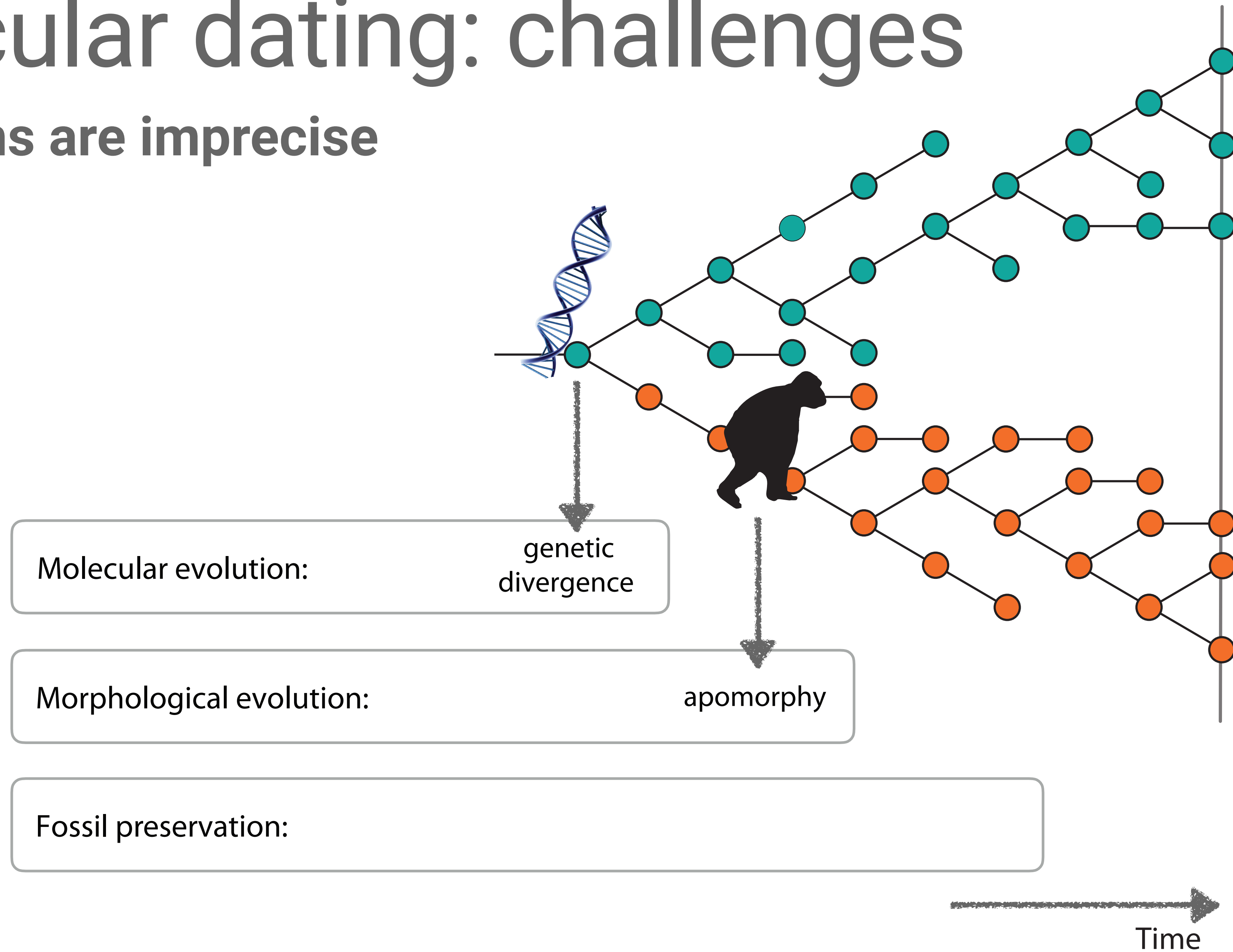
Molecular dating: challenges

Calibrations are imprecise



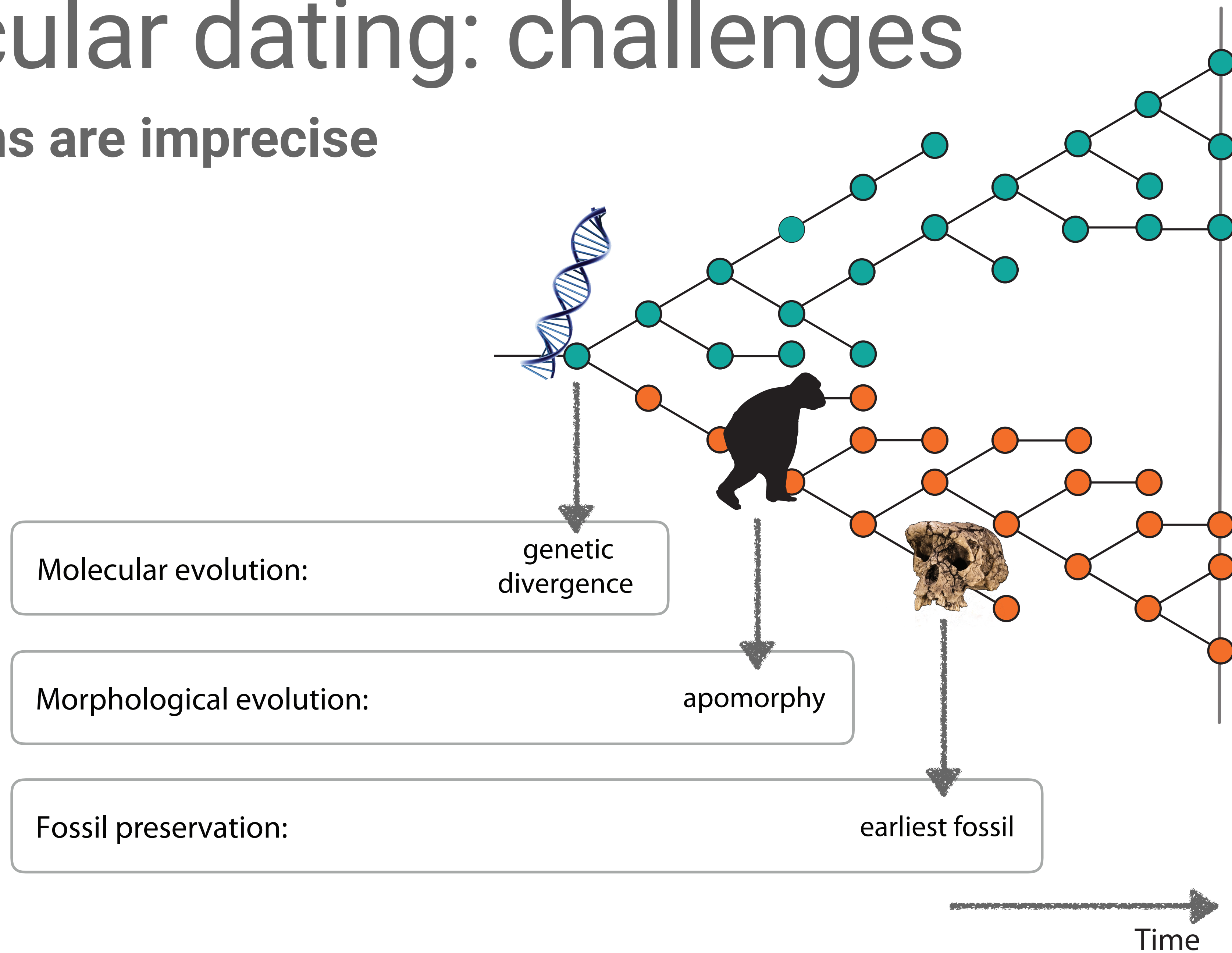
Molecular dating: challenges

Calibrations are imprecise



Molecular dating: challenges

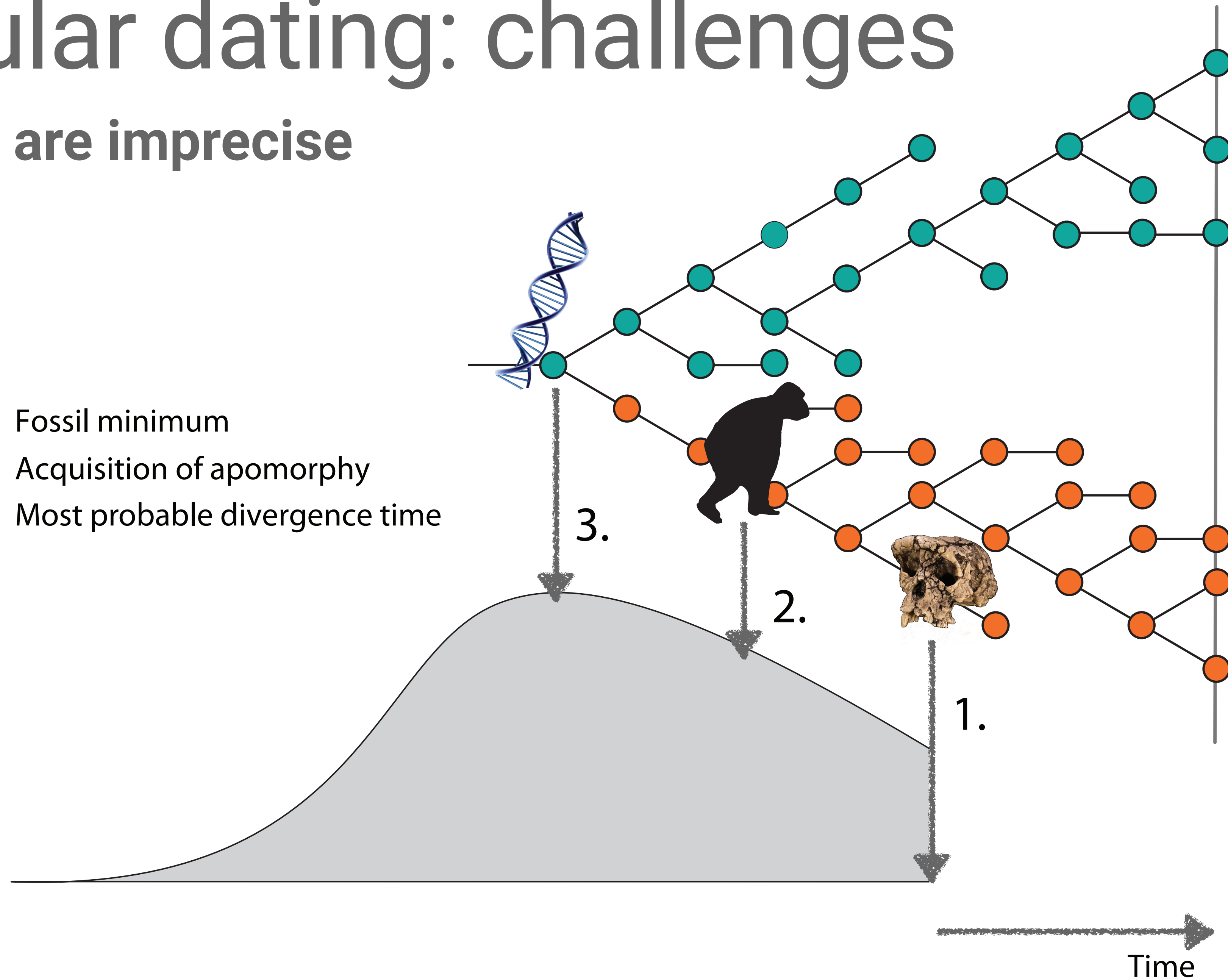
Calibrations are imprecise



Molecular dating: challenges

Calibrations are imprecise

1. Fossil minimum
2. Acquisition of apomorphy
3. Most probable divergence time



Molecular dating: challenges

Summary

1. Rate and time are not fully identifiable

2. The substitution rate varies

3. Calibrations are imprecise

→ we need a flexible statistical framework that deals well with uncertainty!

Bayesian divergence time estimation

We use a Bayesian framework

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model}) P(\text{model})}{P(\text{data})}$$

Diagram illustrating the Bayesian framework components and their relationships:

- likelihood**: Points to $P(\text{data} \mid \text{model})$
- priors**: Points to $P(\text{model})$
- posterior**: Points to $P(\text{model} \mid \text{data})$
- marginal probability of the data**: Points to $P(\text{data})$

Bayesian divergence time estimation

The data

and / or

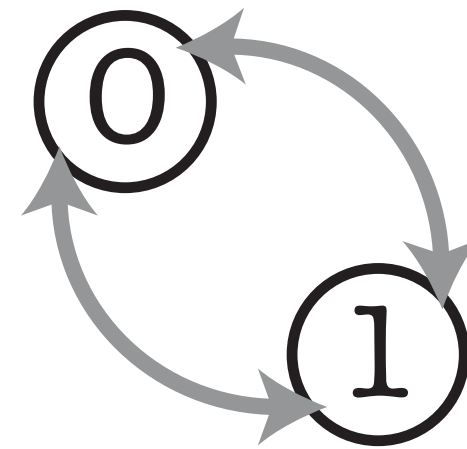
0101... ATTG...
1101... TTGC...
0100... ATTC...



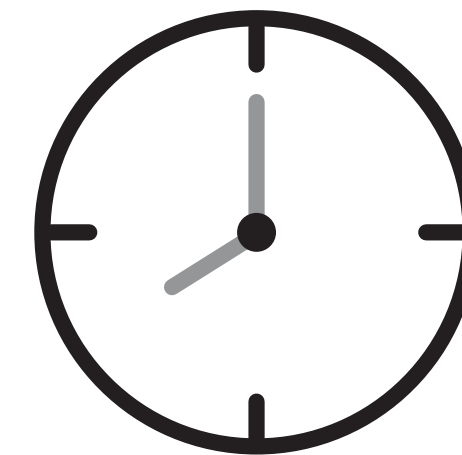
phylogenetics
characters

sample
ages

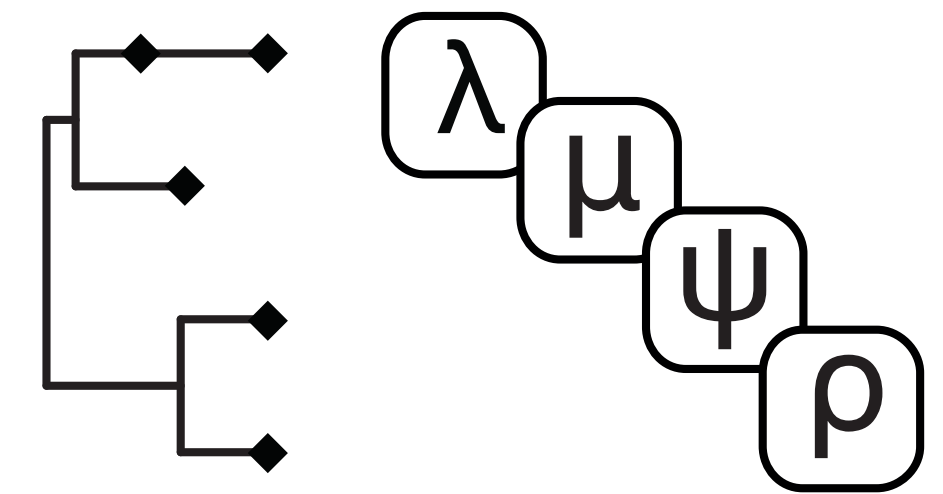
3 model components



substitution
model

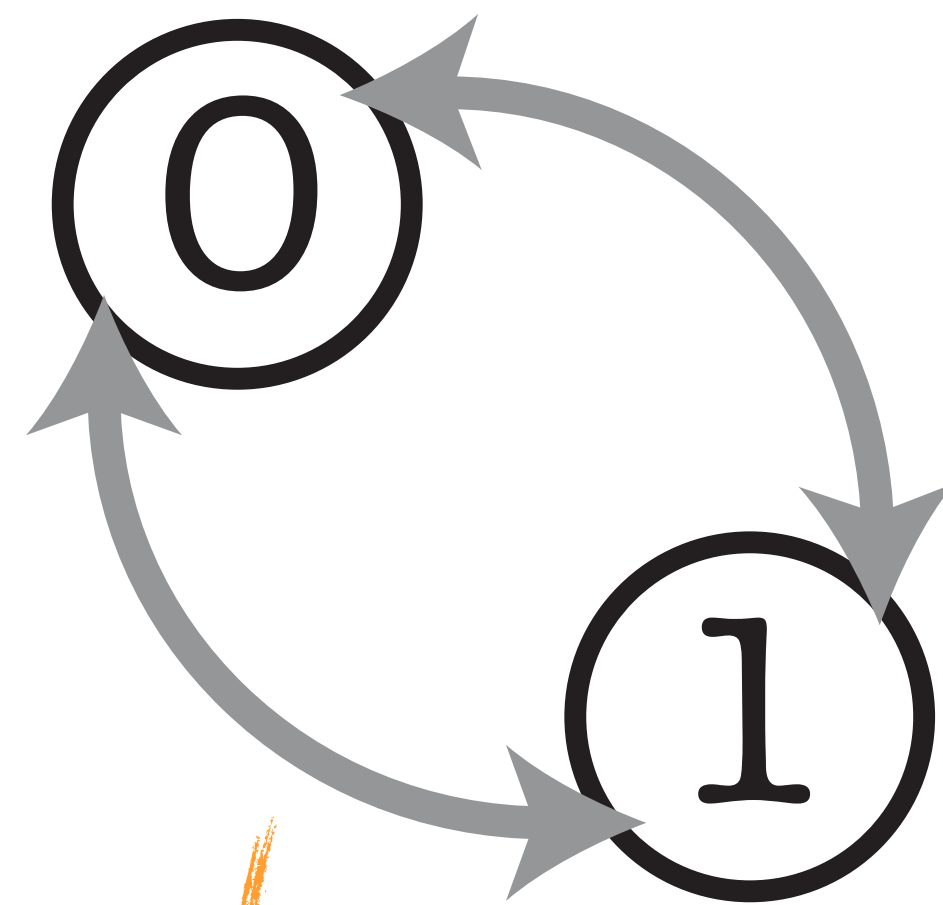


clock
model

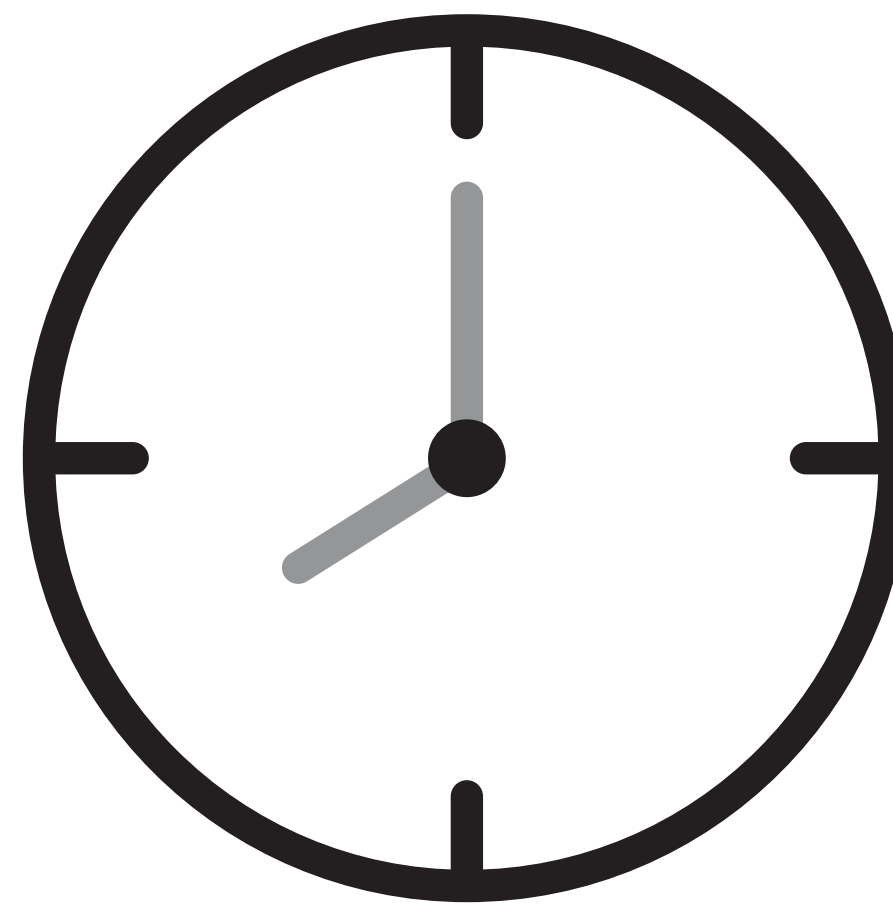


tree and tree
model

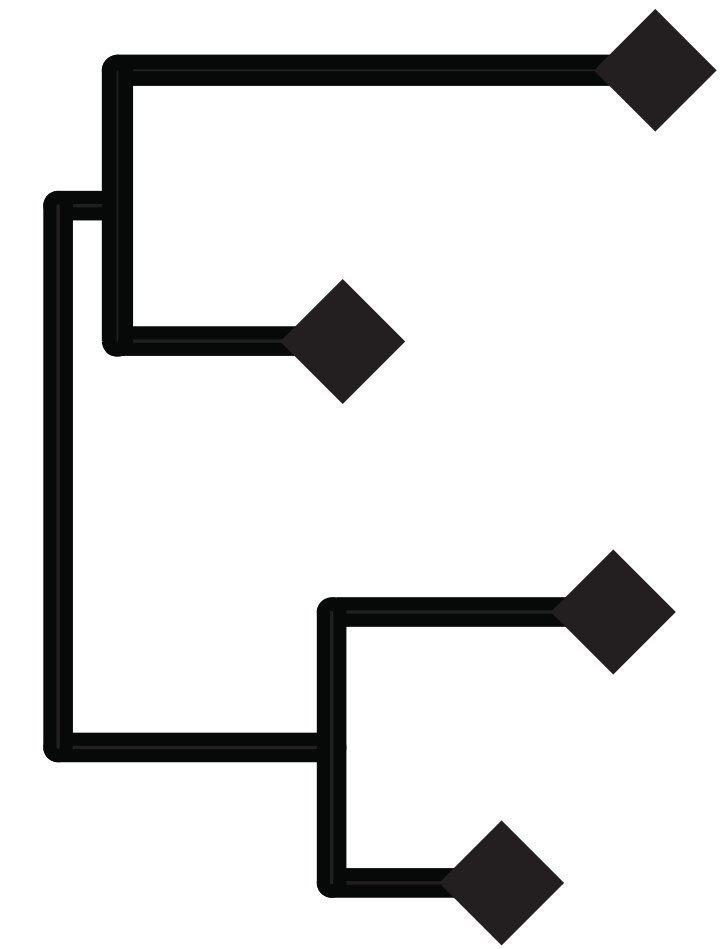
substitution model



clock model

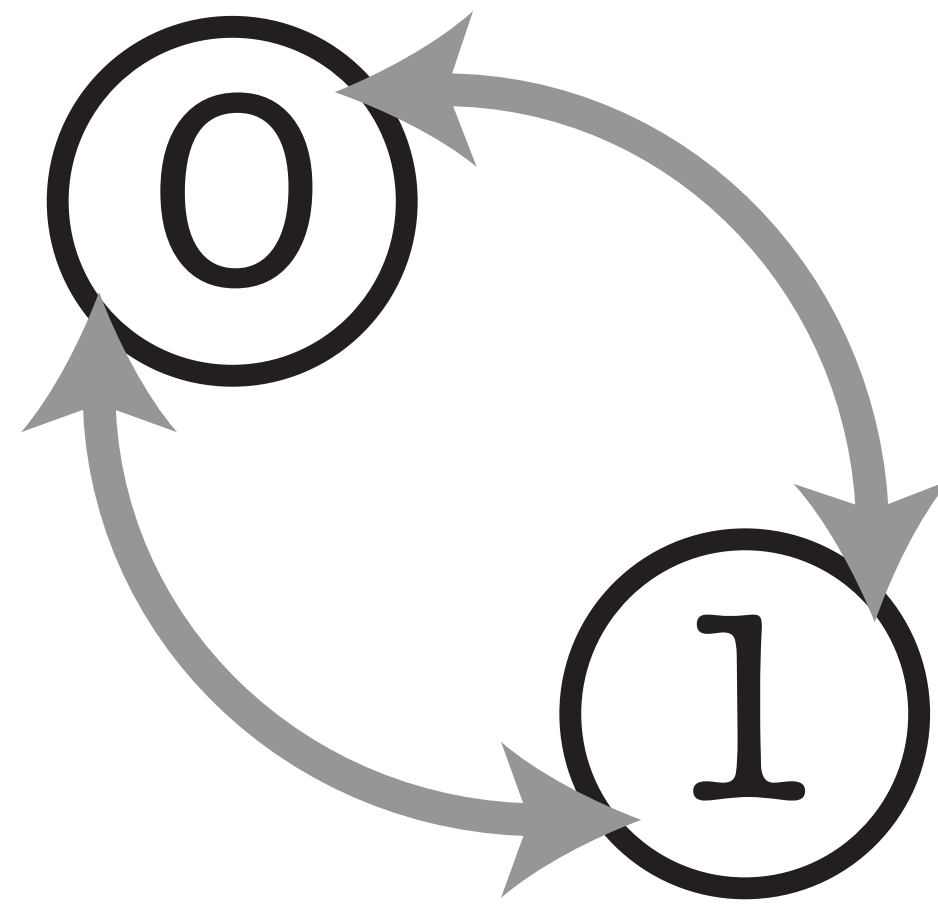


tree model

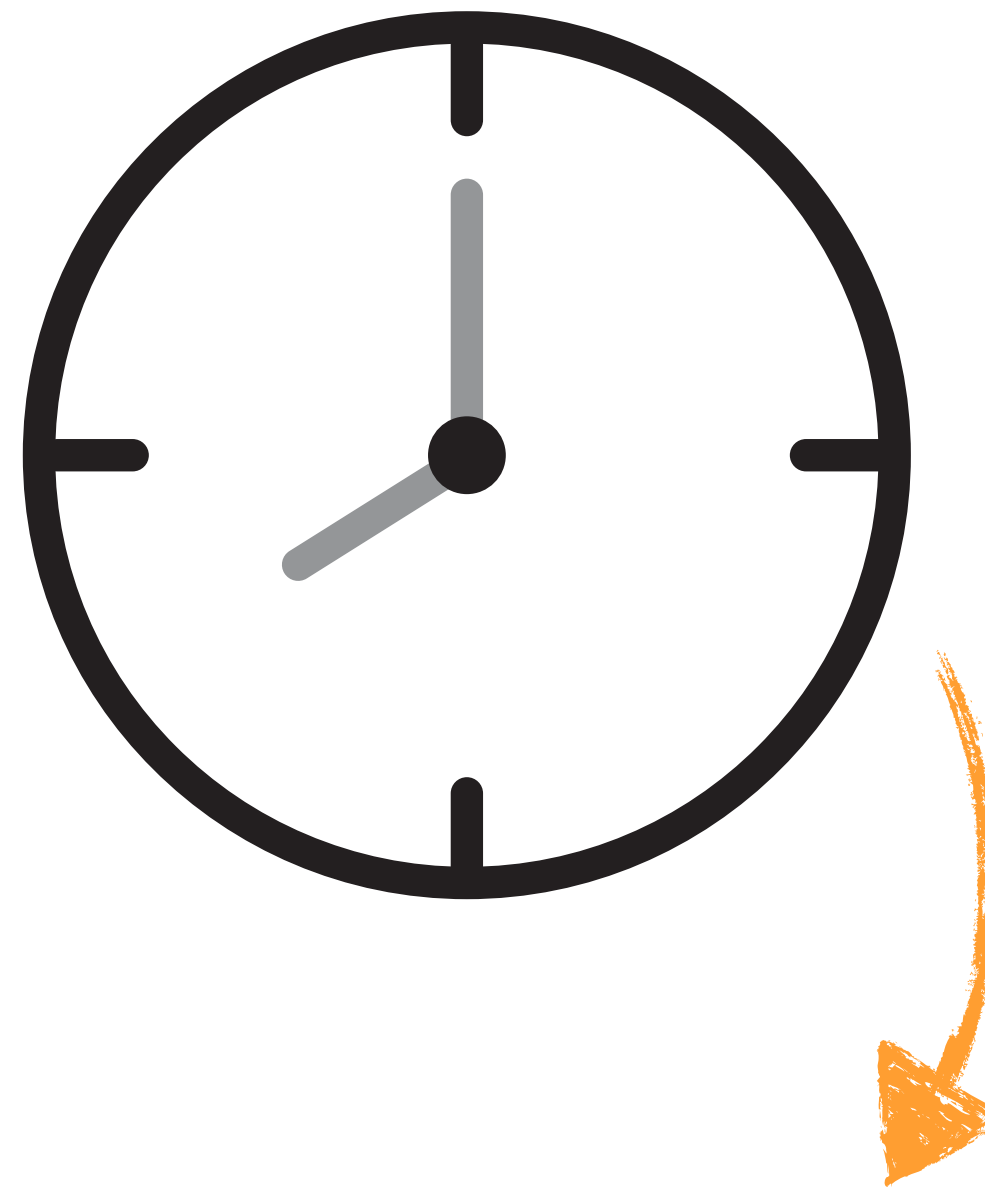


How likely are we to observe a change between character states? e.g., $A \rightarrow T$

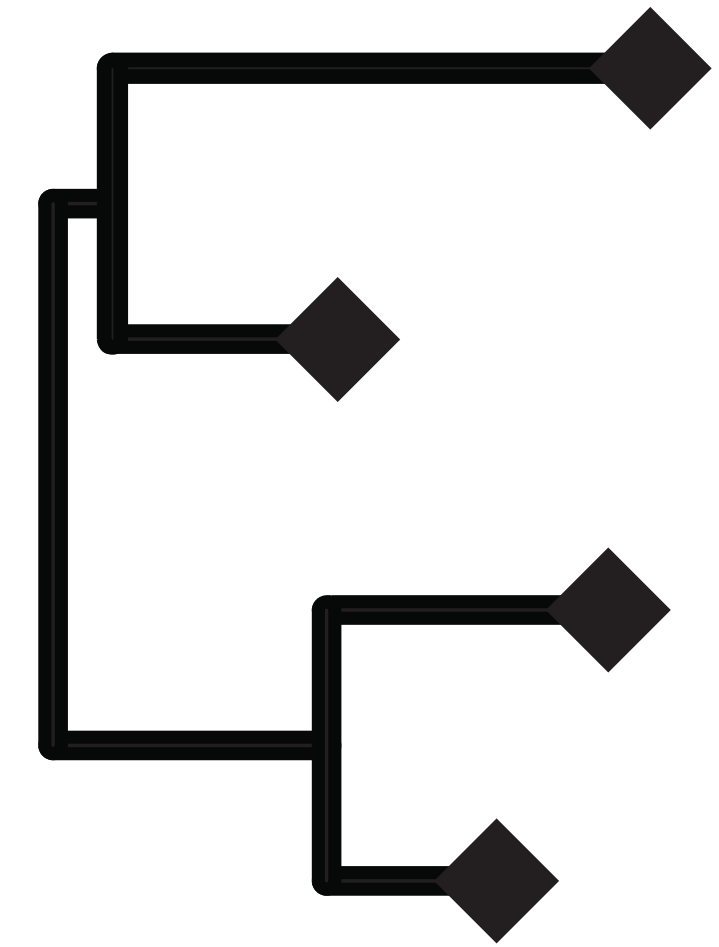
substitution model



clock model

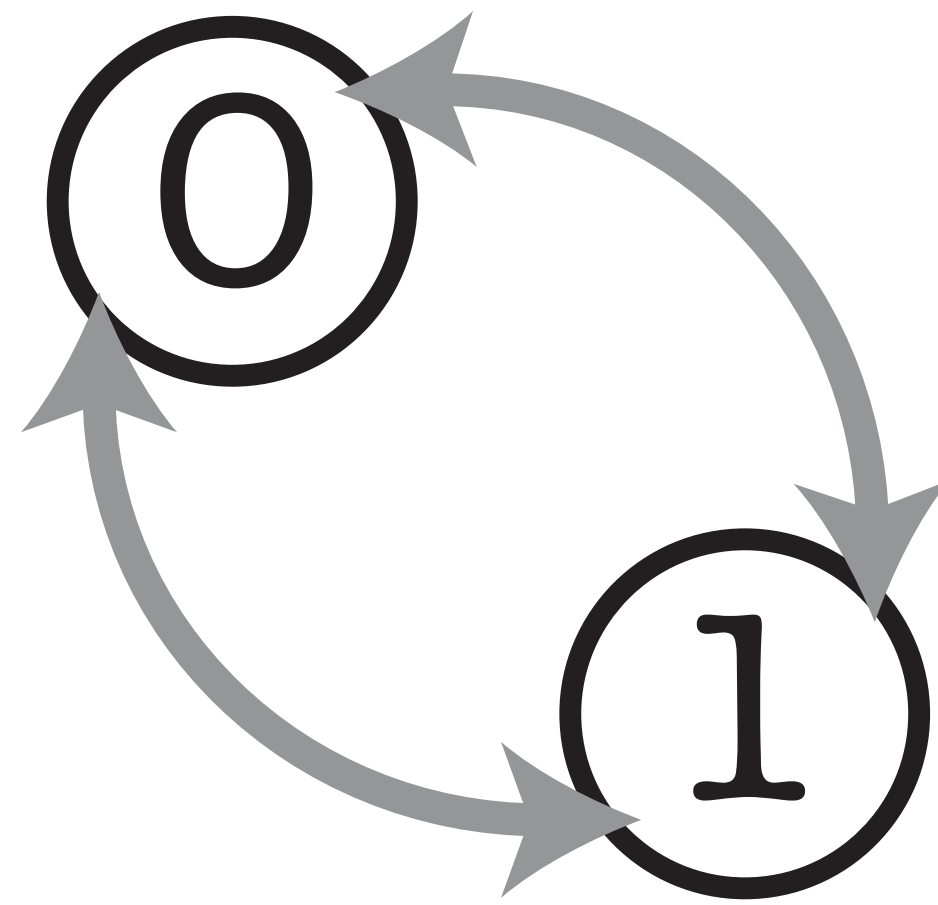


tree model

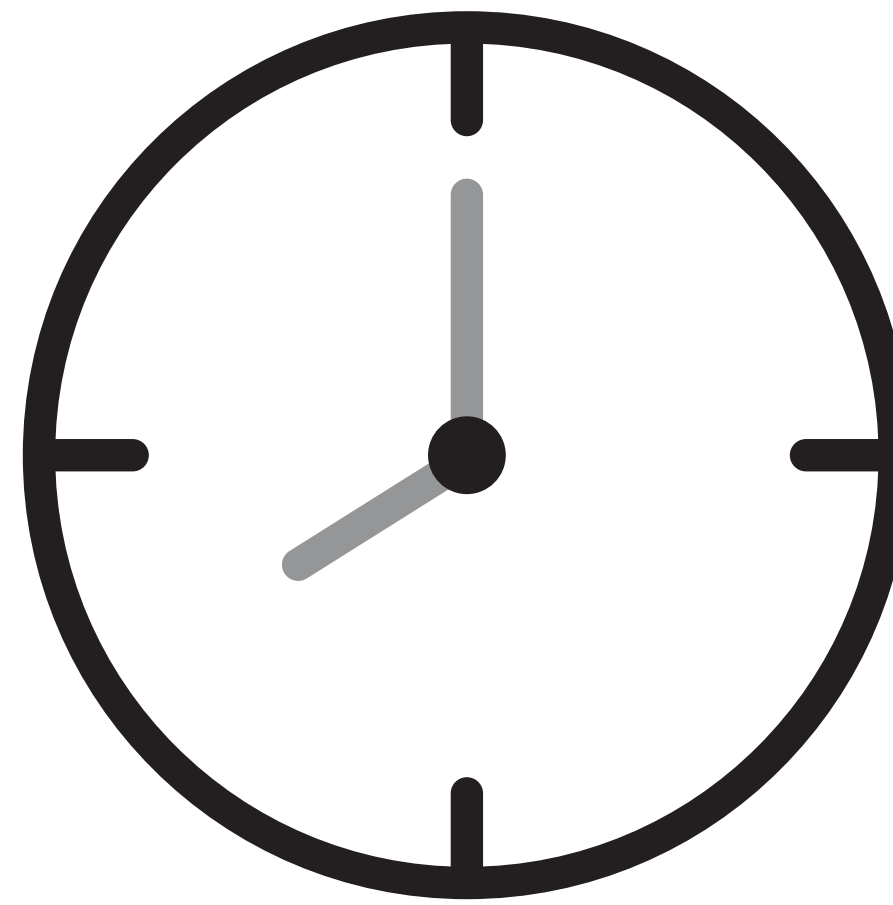


How have rates of evolution varied
(or not) across the tree?

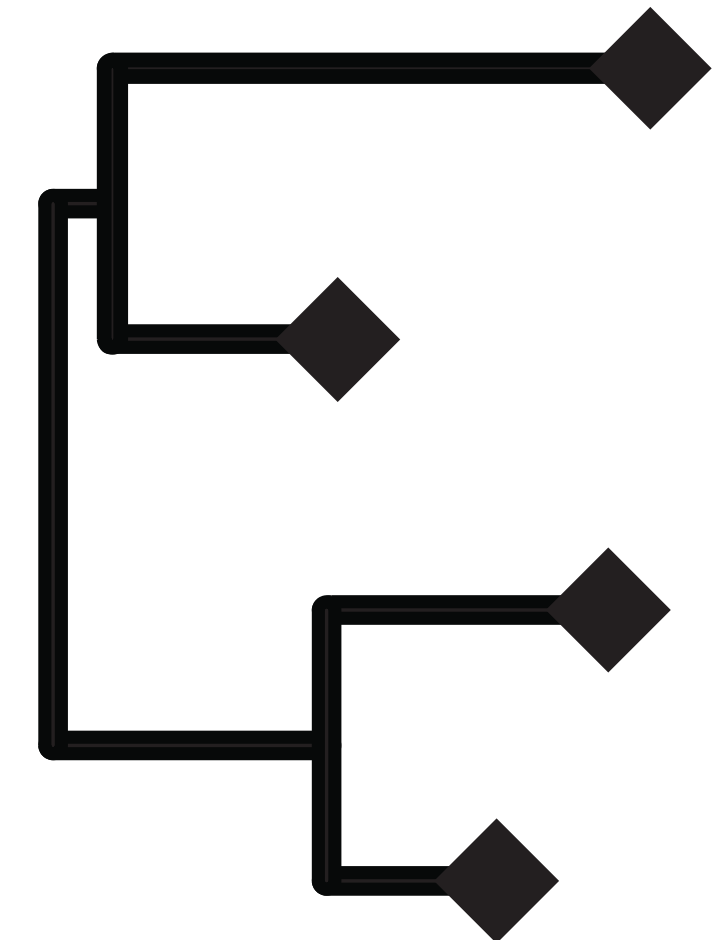
substitution model



clock model



tree model



How have species originated, gone extinct and been sampled through time?

Bayesian divergence time estimation

posterior

$$P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \begin{array}{c} \text{0} \\ \text{1} \end{array} \text{clock} \mid \begin{array}{l} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \text{data}\right) =$$

likelihood

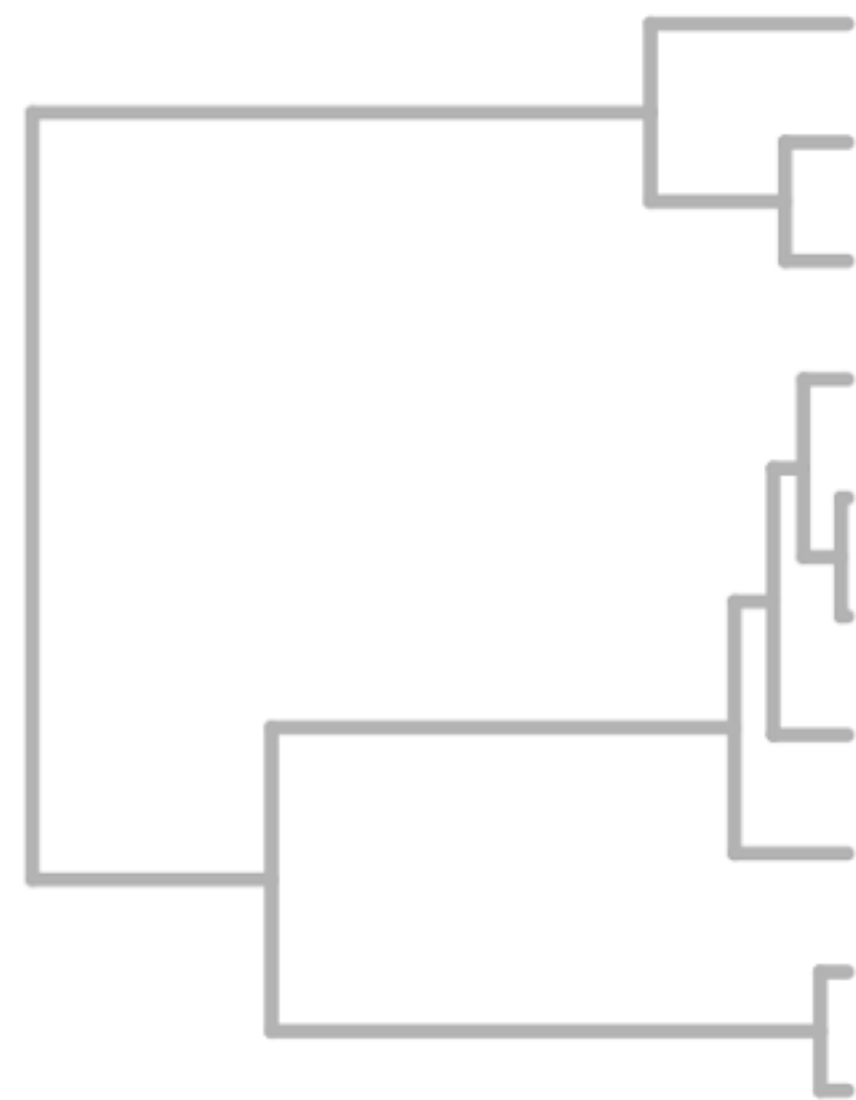
probability of the
time tree

priors

$$\frac{P\left(\begin{array}{l} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \begin{array}{c} \text{0} \\ \text{1} \end{array} \text{clock}\right) P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \text{data}\right) P\left(\begin{array}{c} \lambda \\ \mu \\ \psi \\ \rho \end{array}\right) P\left(\begin{array}{c} \text{0} \\ \text{1} \end{array}\right) P(\text{clock})}{P\left(\begin{array}{l} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \text{data}\right)}$$

marginal pr of the data

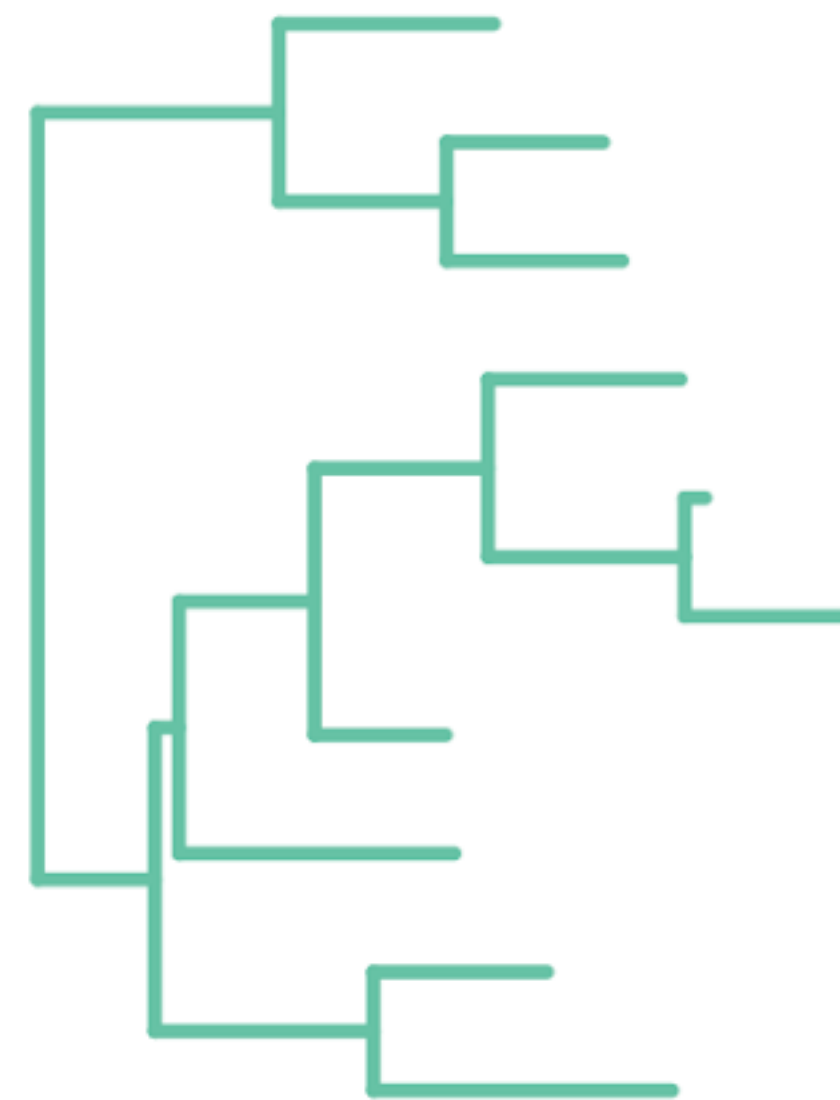
Calculating the likelihood



Time (year)

Prior

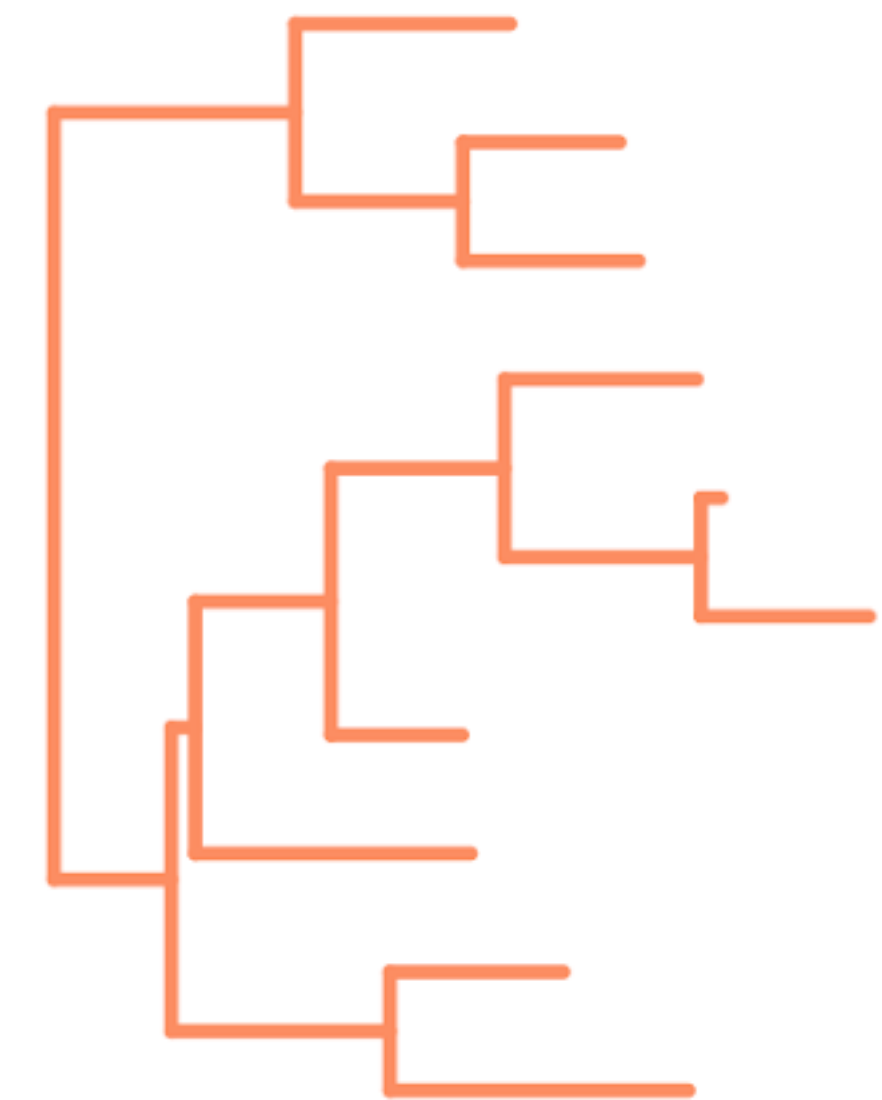
X



Clock rate (subs/site/year)

Prior

=



Genetic distance (subs/site)

Likelihood

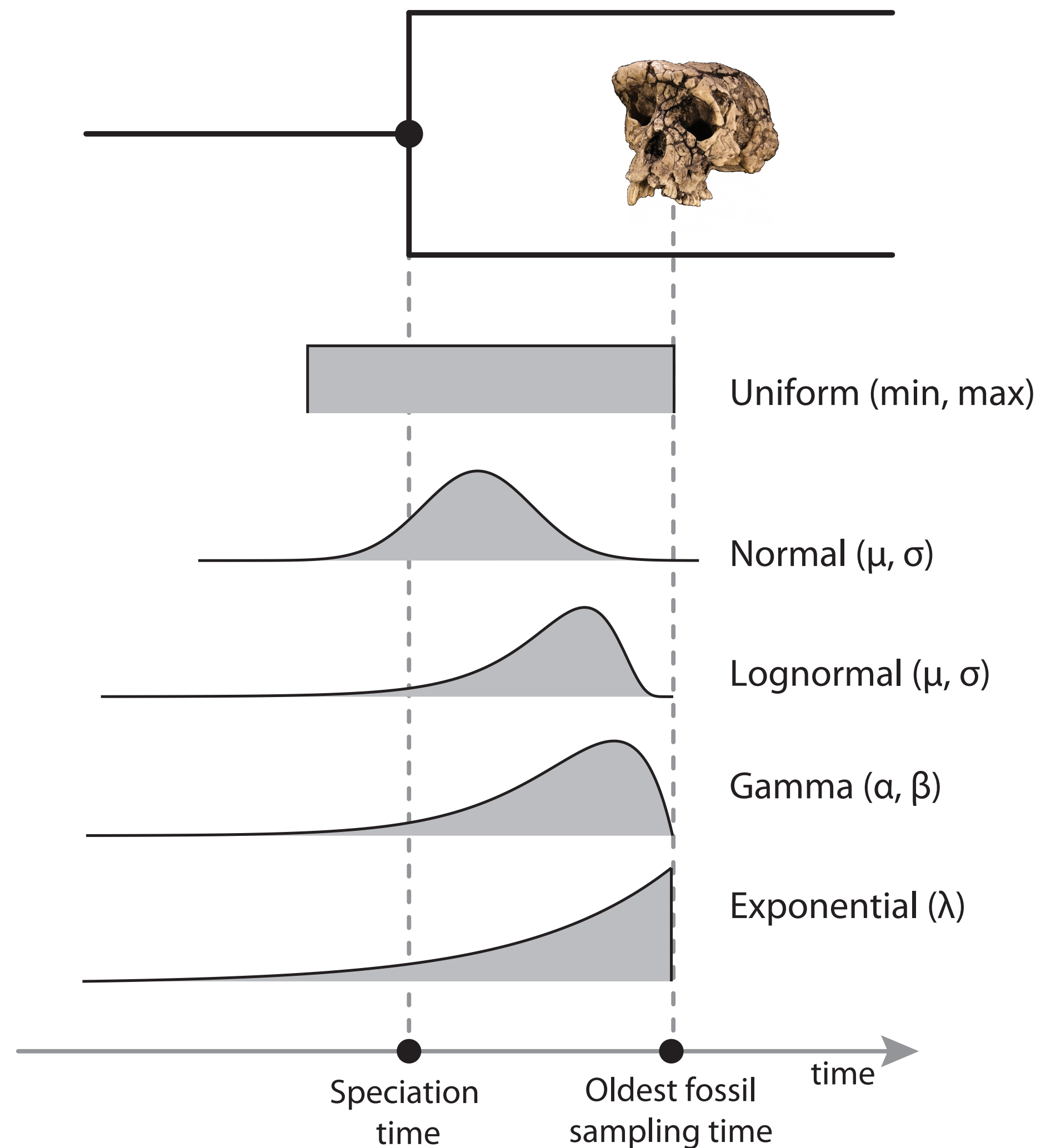


Based on the calibration times we can estimate the rate over time



Once we have the rate we can transform evolutionary rates in genetic distance

Node dating

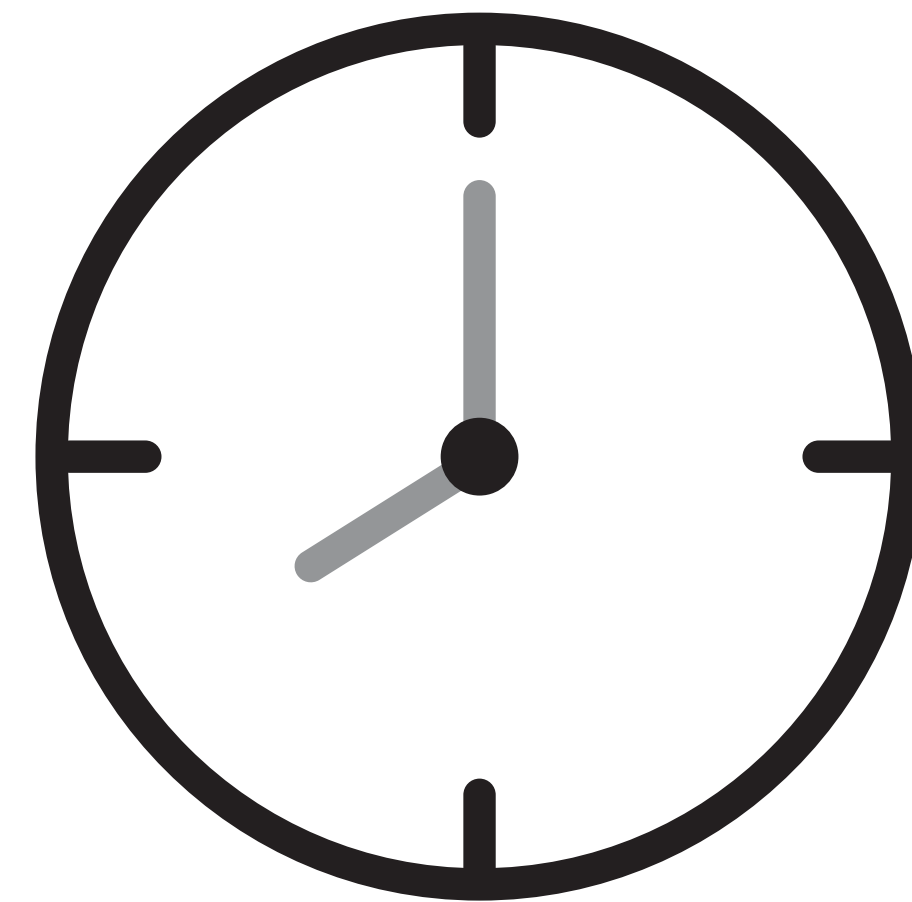


We can use a **calibration density** to constrain internal node ages

We typically use a **birth-death process** model to describe the tree generating process

The clock model describes how evolutionary rates vary (or not) across the tree

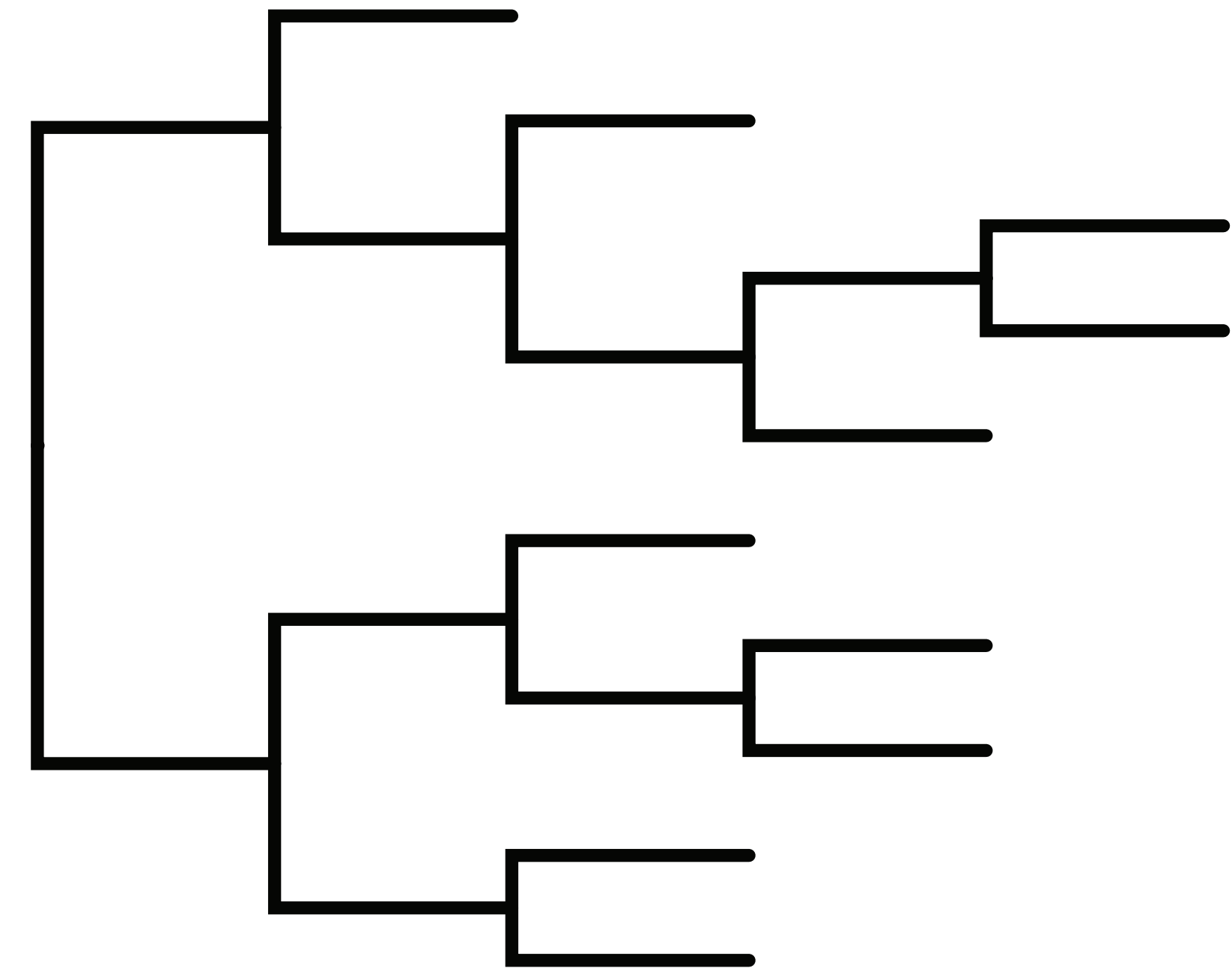
clock model



The strict / constant molecular clock model

Assumptions

- The substitution rate is constant over time
- All lineages share the same rate



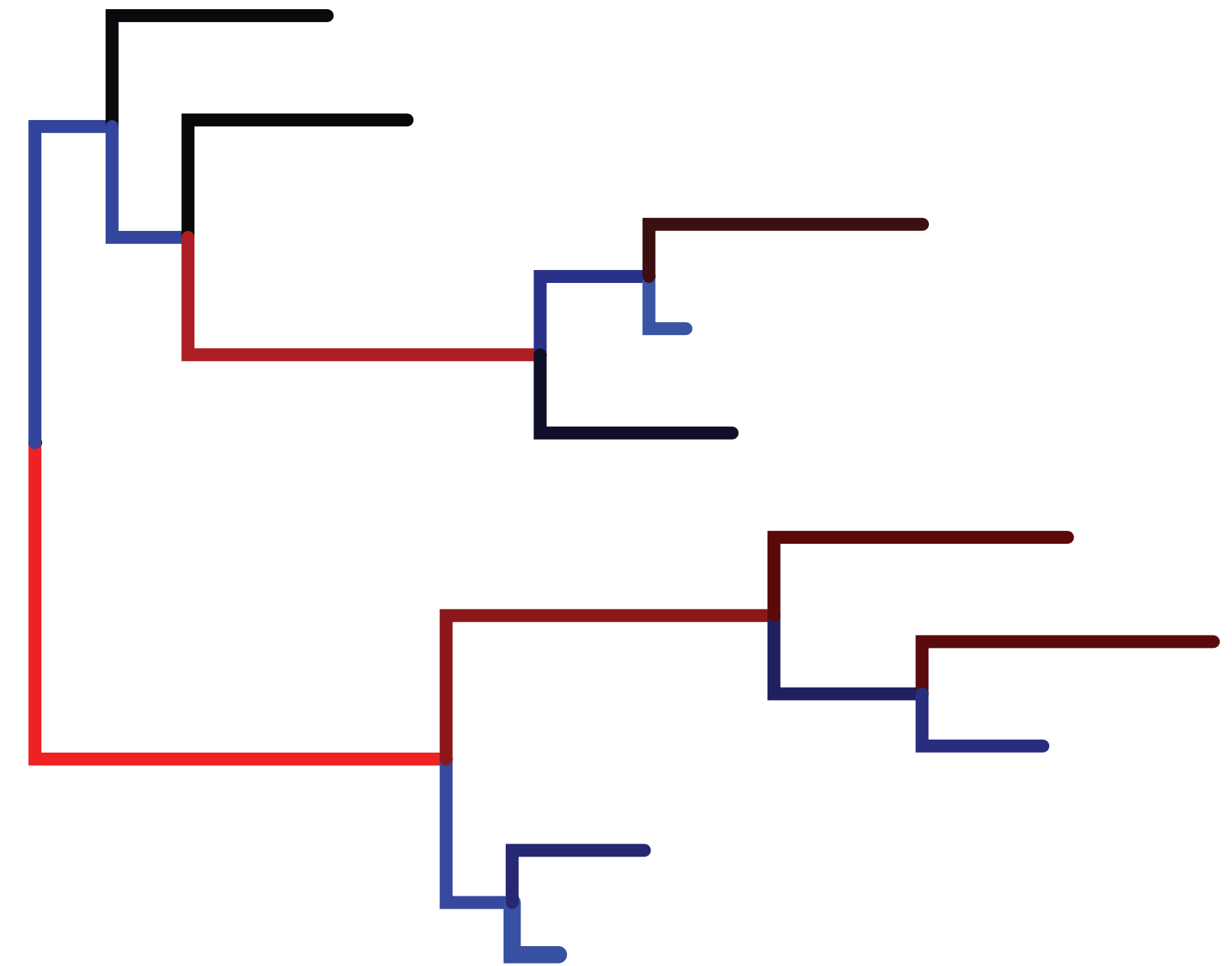
branch length = substitution rate

low  high

Relaxed clock models

Assumptions

- Lineage-specific rates
- The rate assigned to each branch is drawn from some underlying distribution



branch length = substitution rate
low  high

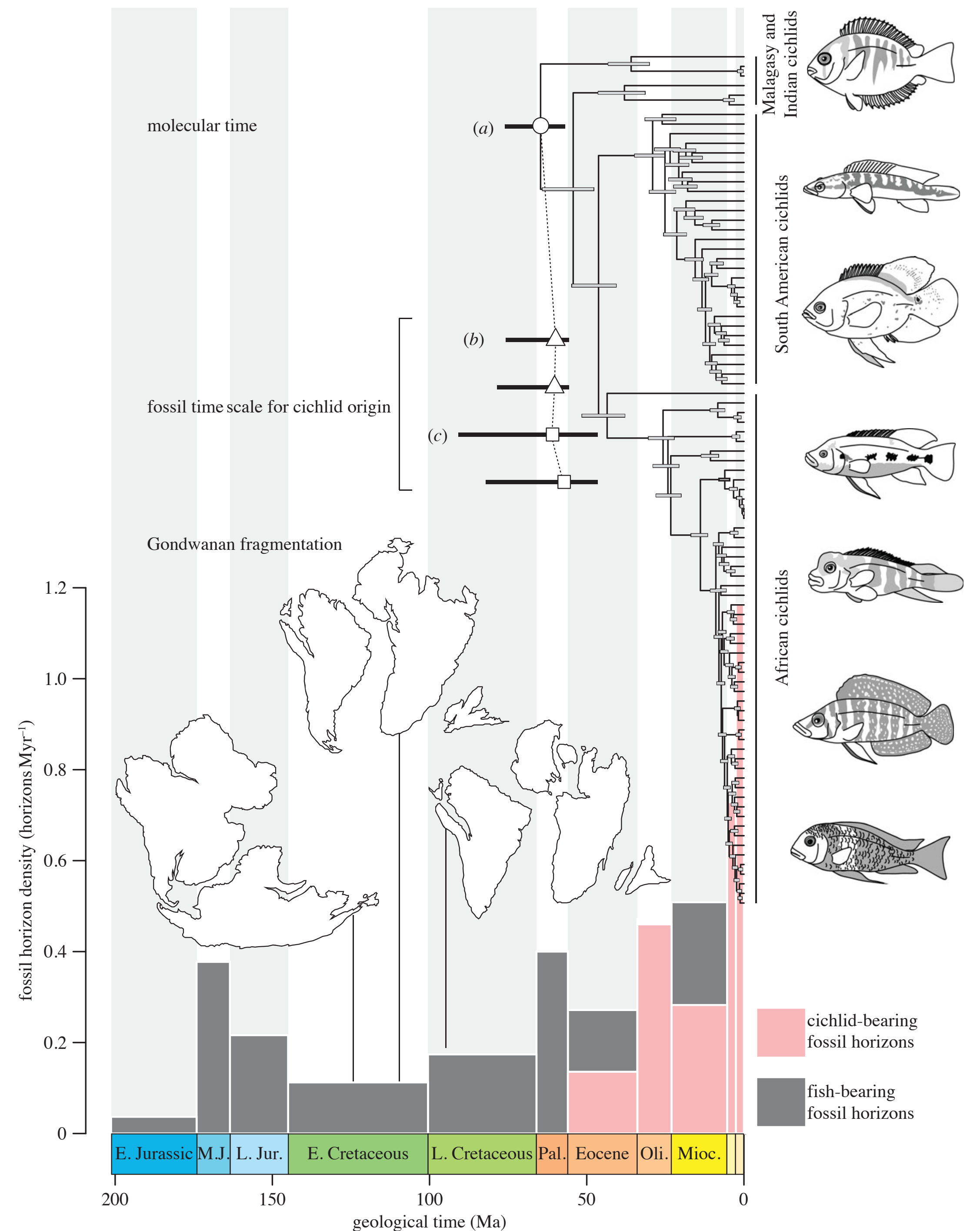
There are many different clock models

- Strict clock
- Uncorrelated or independent clock (= the favourite)
- Autocorrelated clock
- Local clocks
- Mixture models

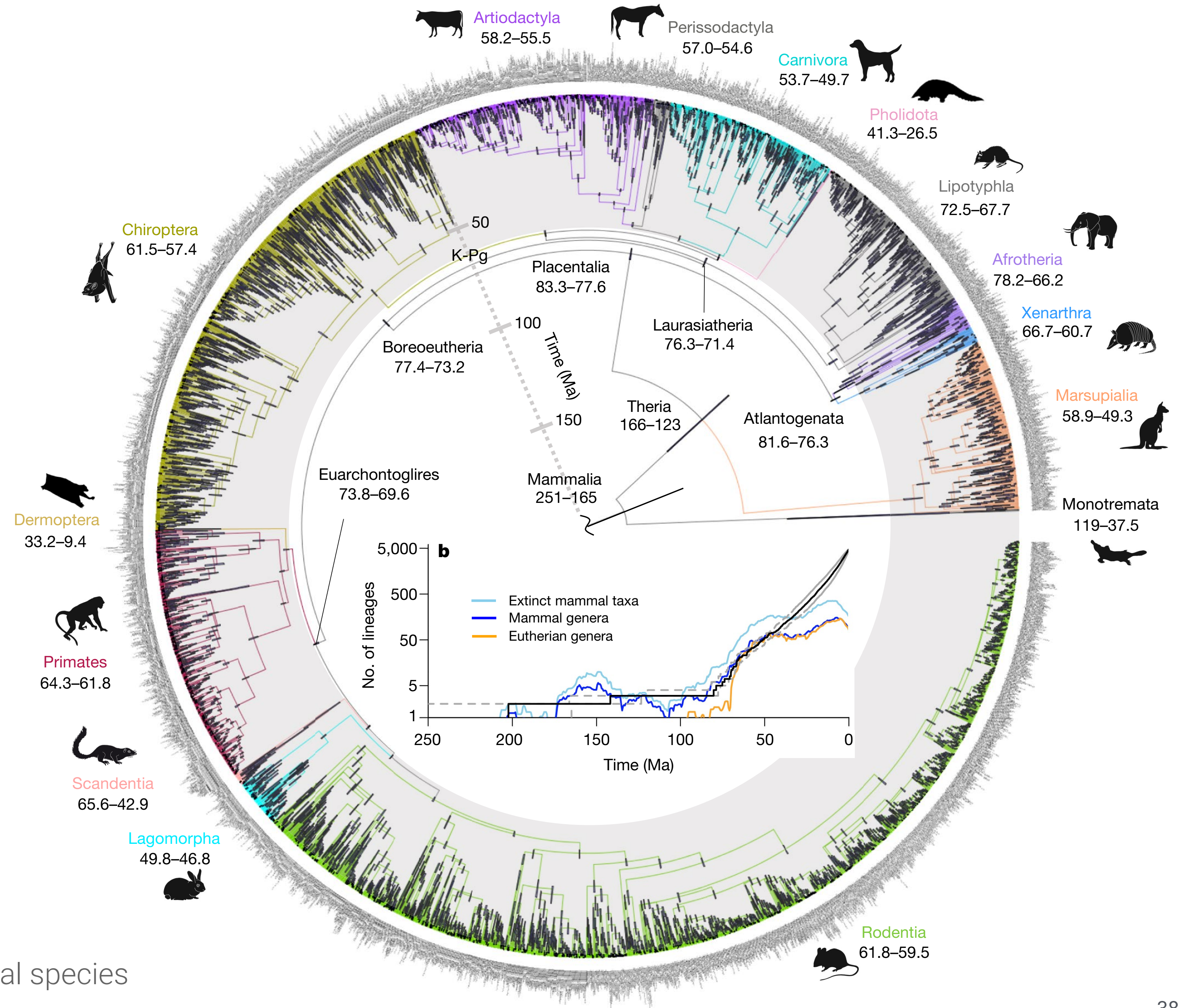
Applications of node dating

Hypothesis testing: did the break up of Gondwana drive the radiation of cichlids?

Image adapted from *Friedmann et al.* ([2013](#))

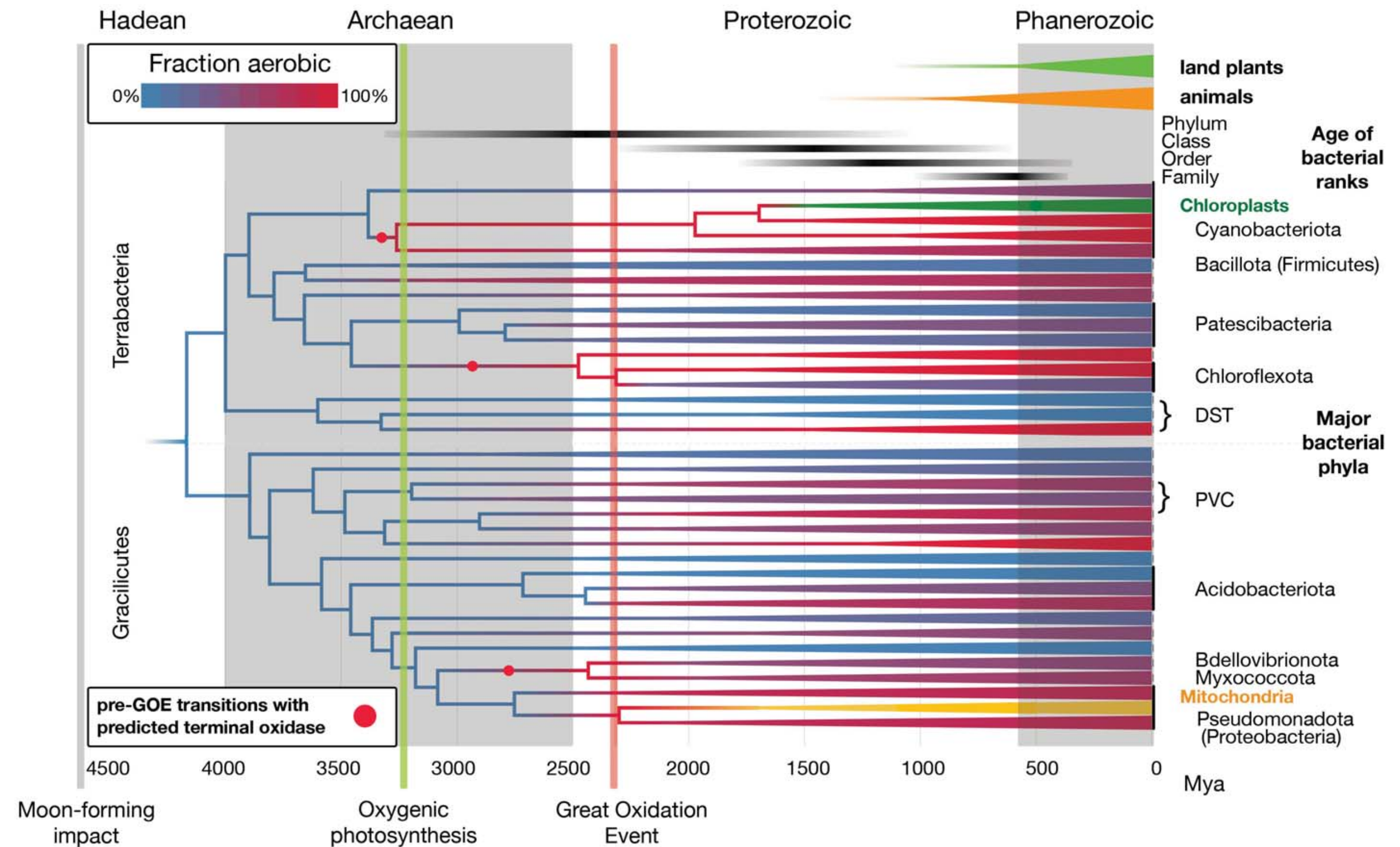


Very large trees
can only be
time-calibrated
using a node
dating
approach



Álvarez-Carretero et al. (2021) — 4,705 mammal species

Dating with sparse calibration information

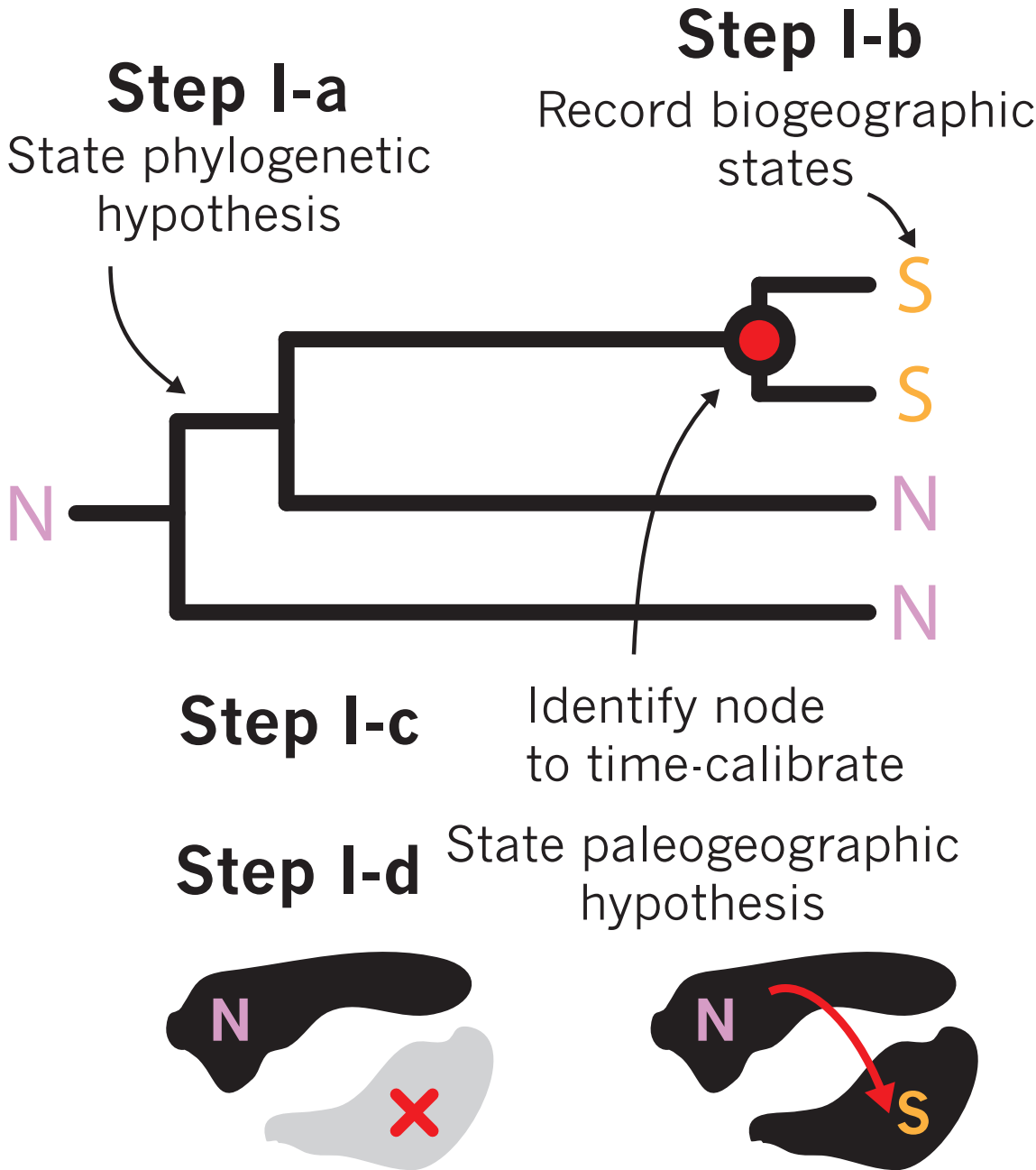


Davín et al. (2025)

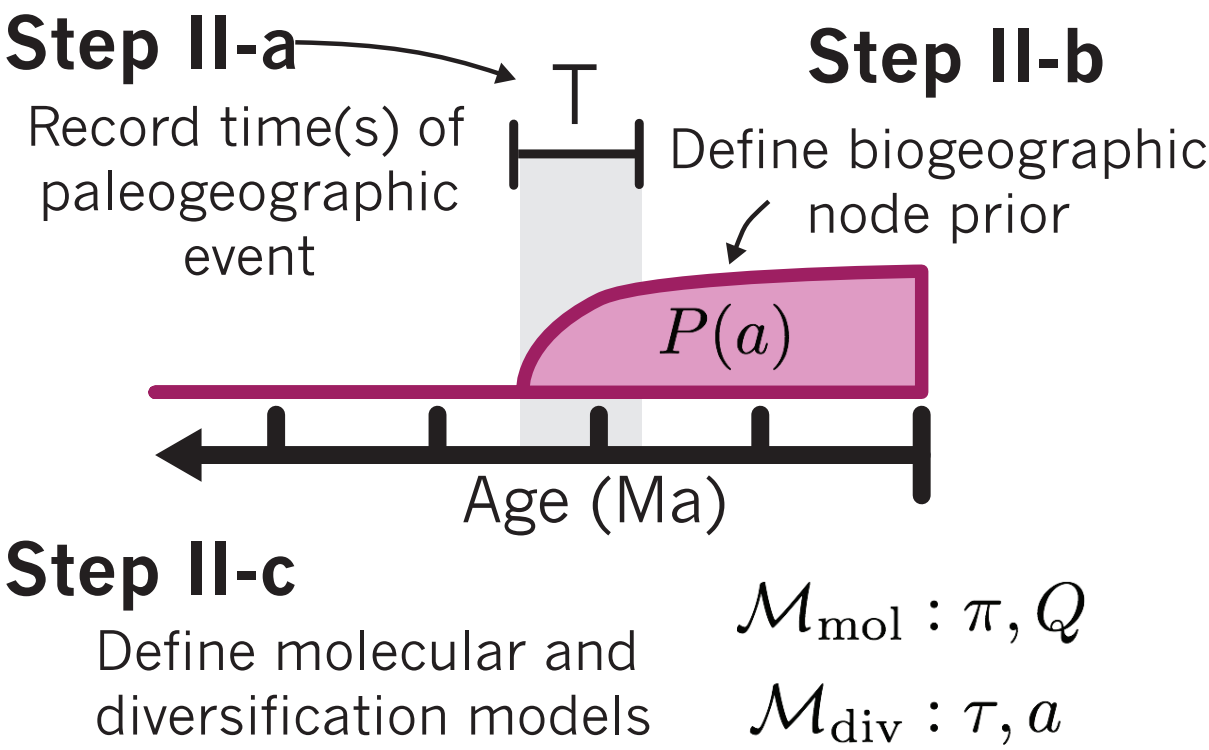
A geological timescale for bacterial evolution, calibrated using atmospheric oxygenation and the spread of aerobic metabolism

Biogeographic calibrations

Step I: Justification



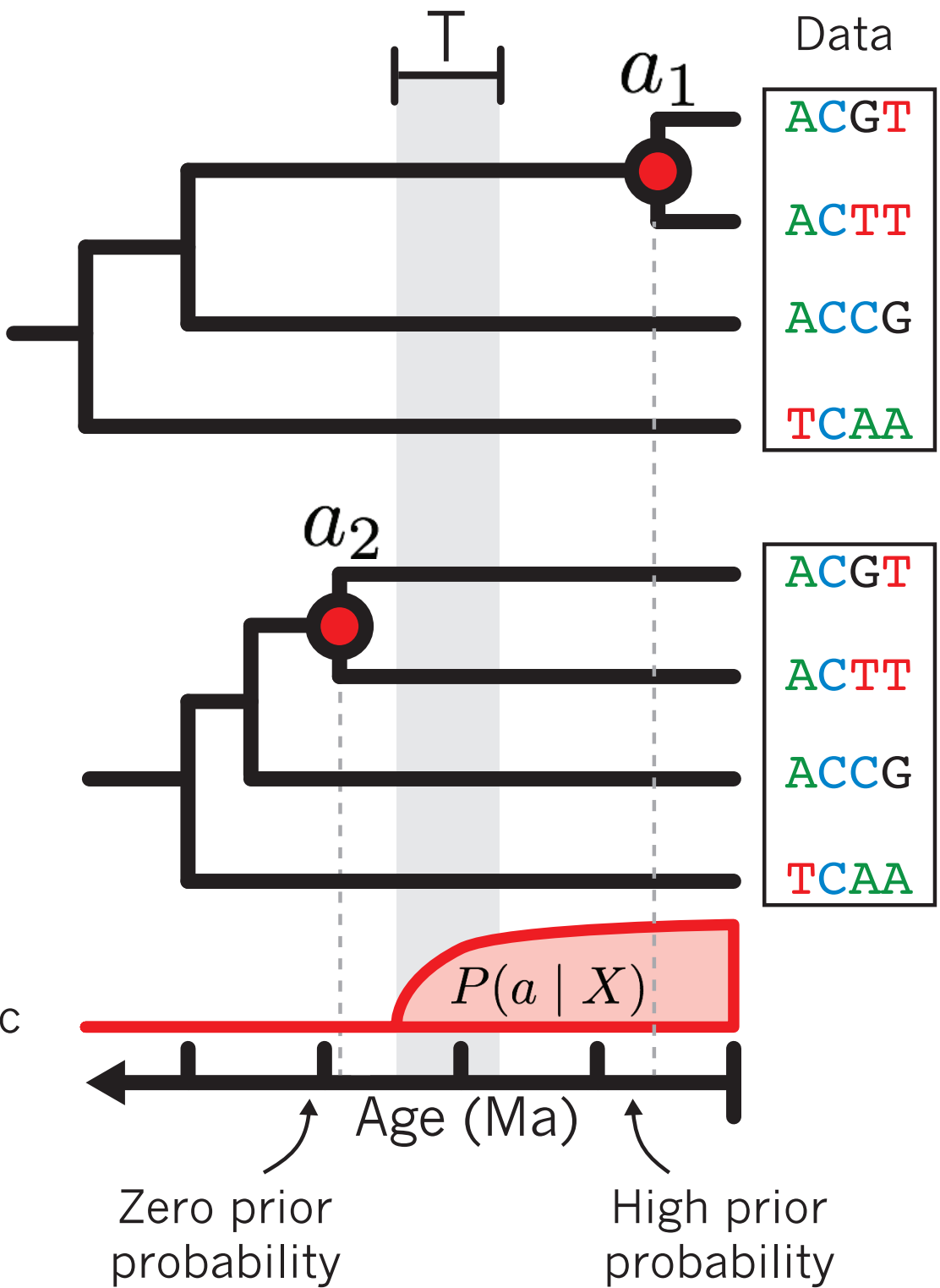
Step II: Specification



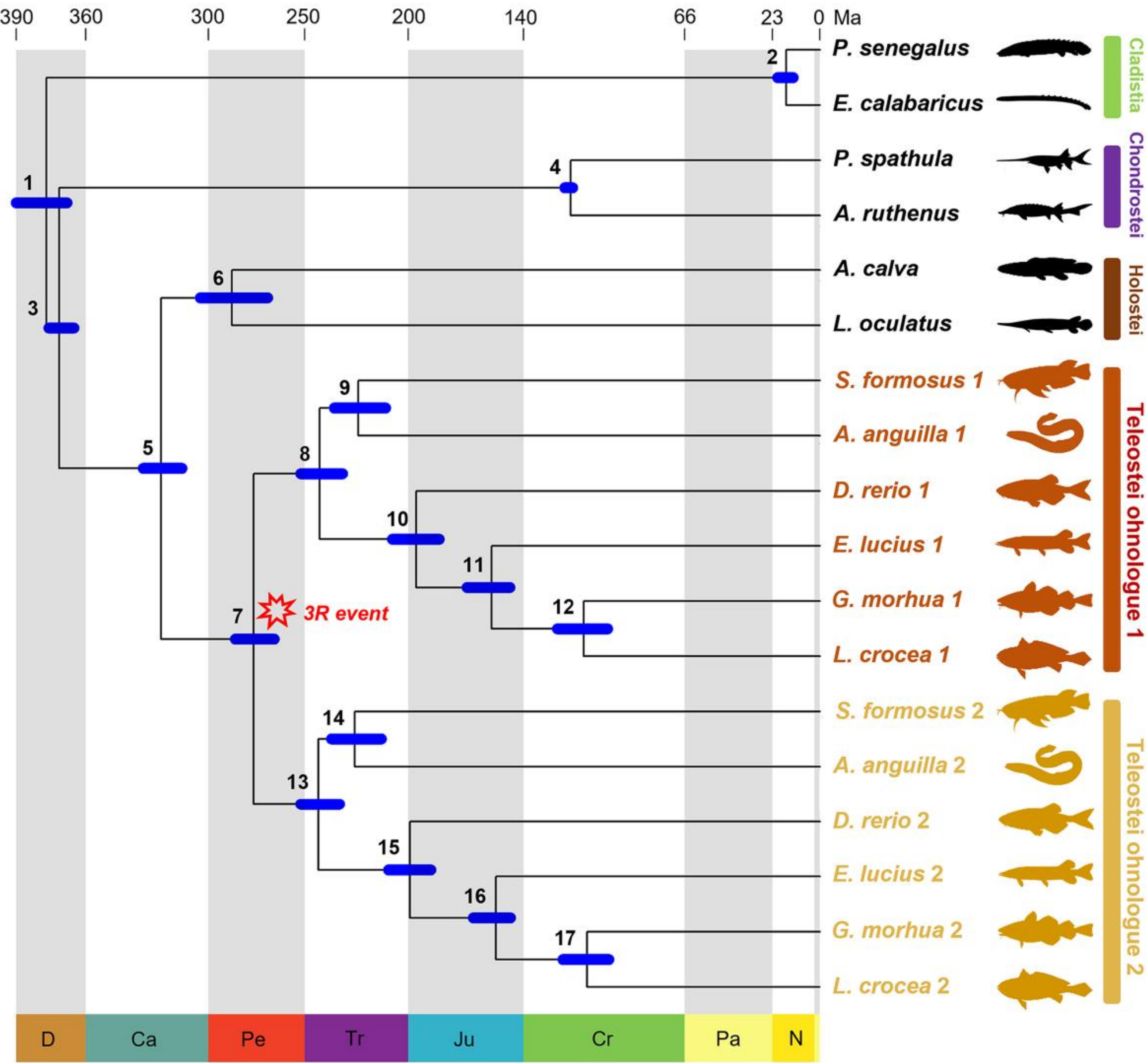
Step III: Estimation

Step III-a

Estimate posterior ages from molecular data and node prior

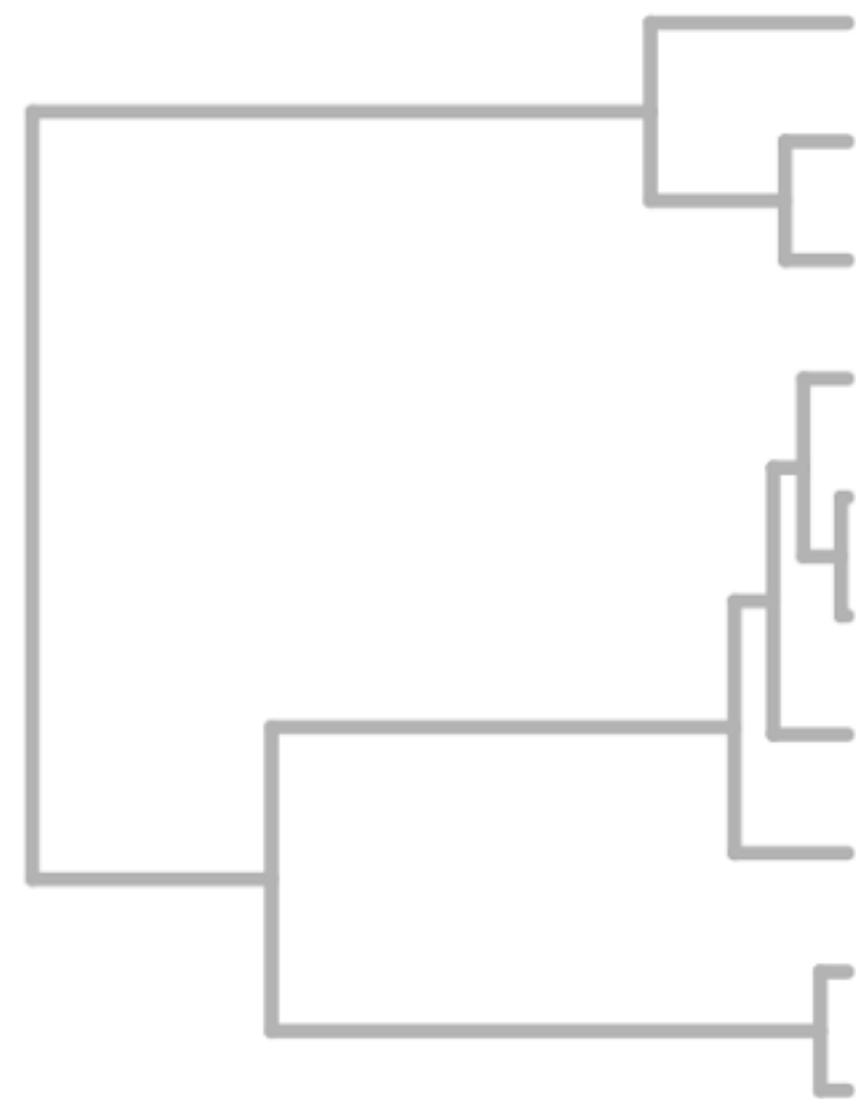


Dating gene or genome duplication events



Álvarez-Carretero et al. (2021) — 4,705 mammal species

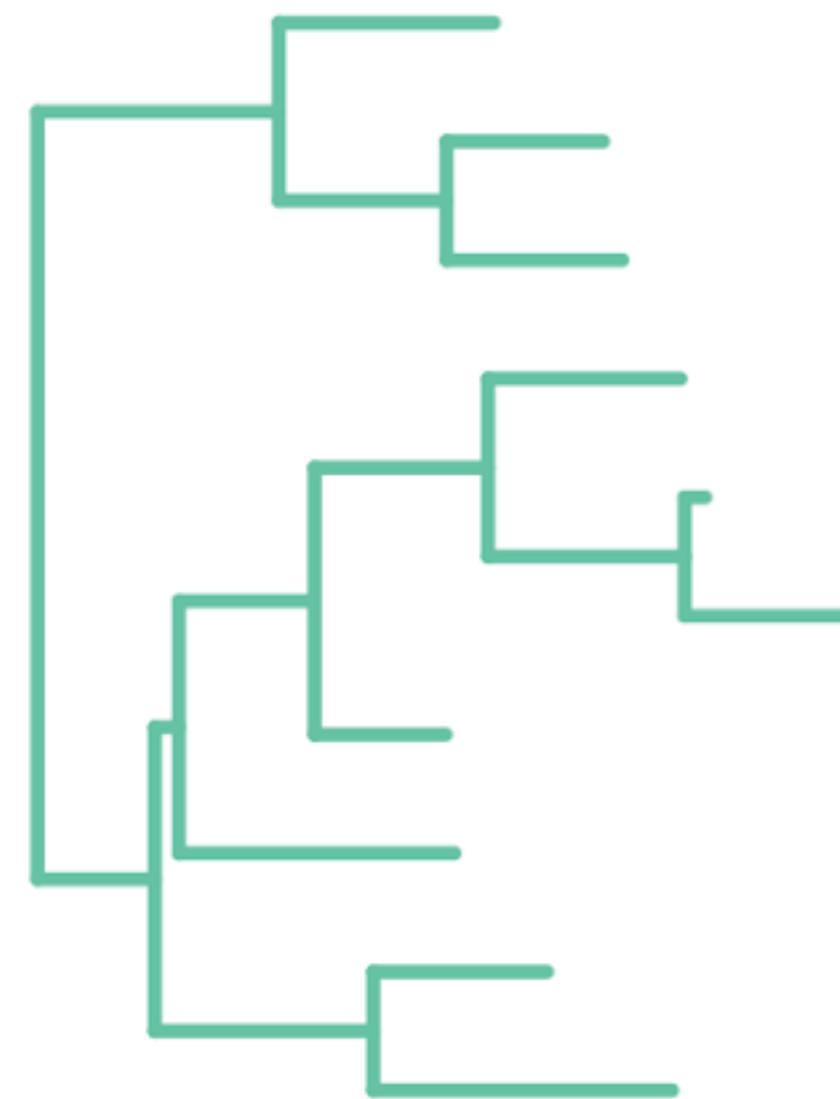
Times and rates are not fully identifiable!



Time (year)

Prior

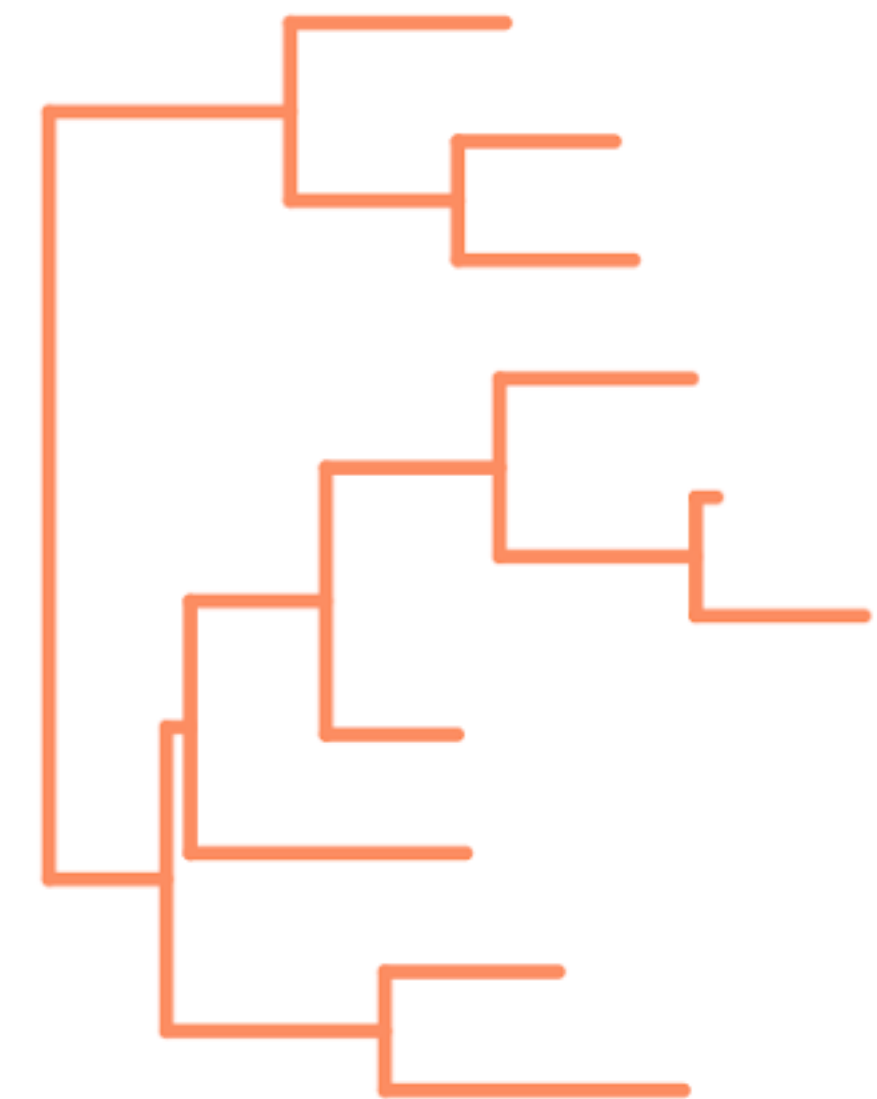
X



Clock rate (subs/site/year)

Prior

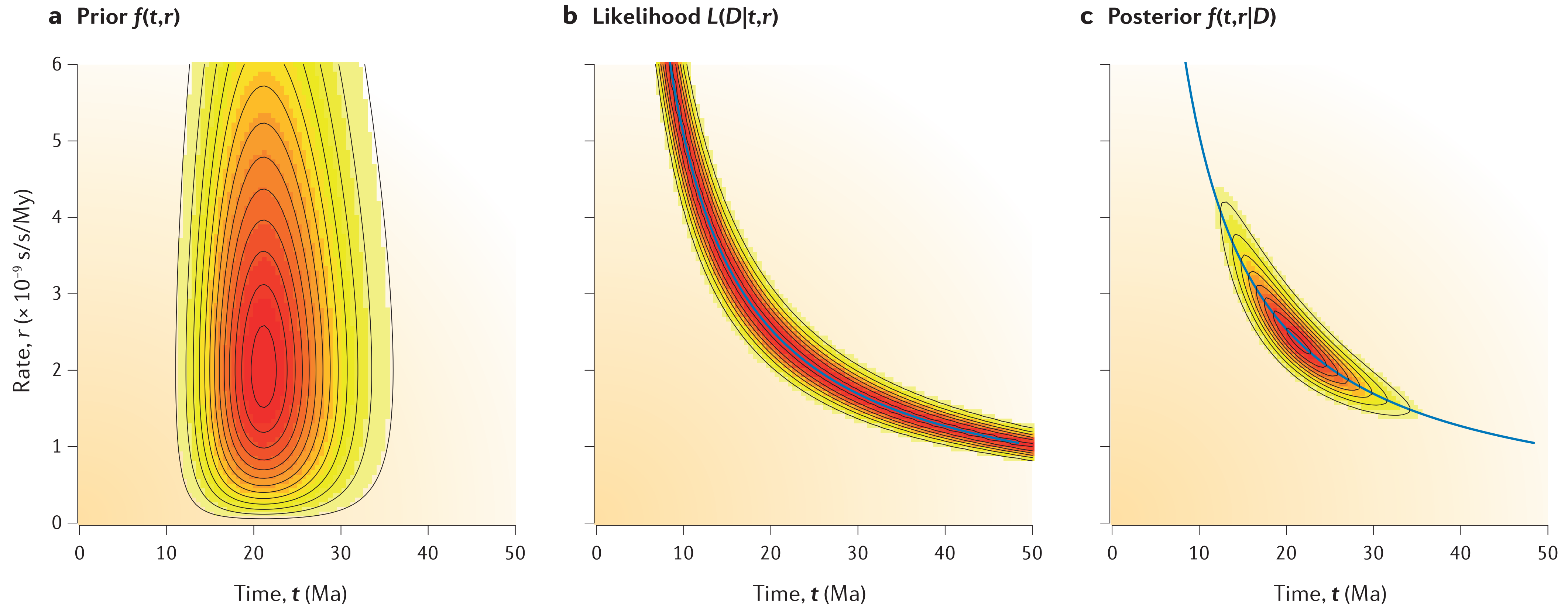
=



Genetic distance (subs/site)

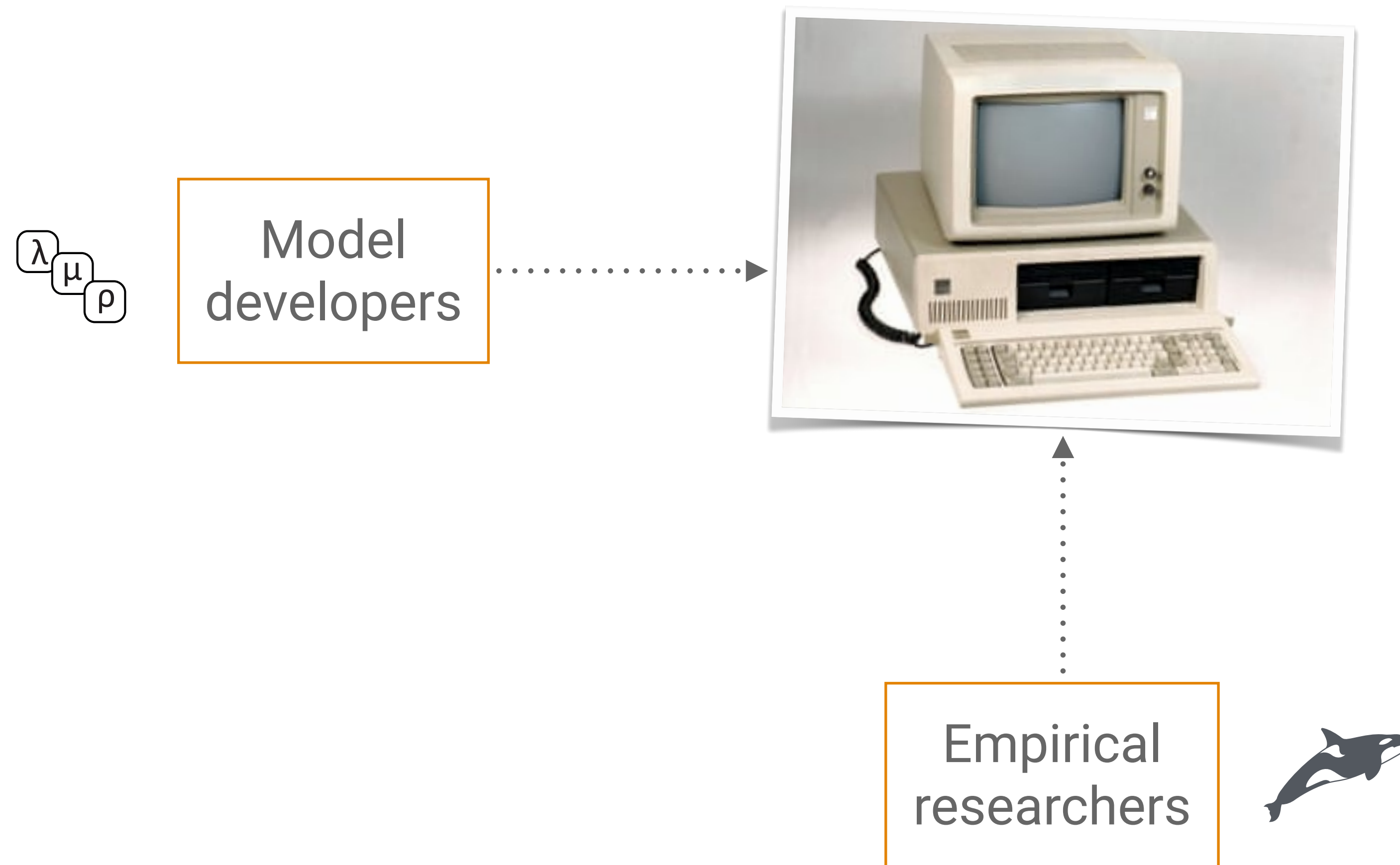
Likelihood

The priors will always influence the results



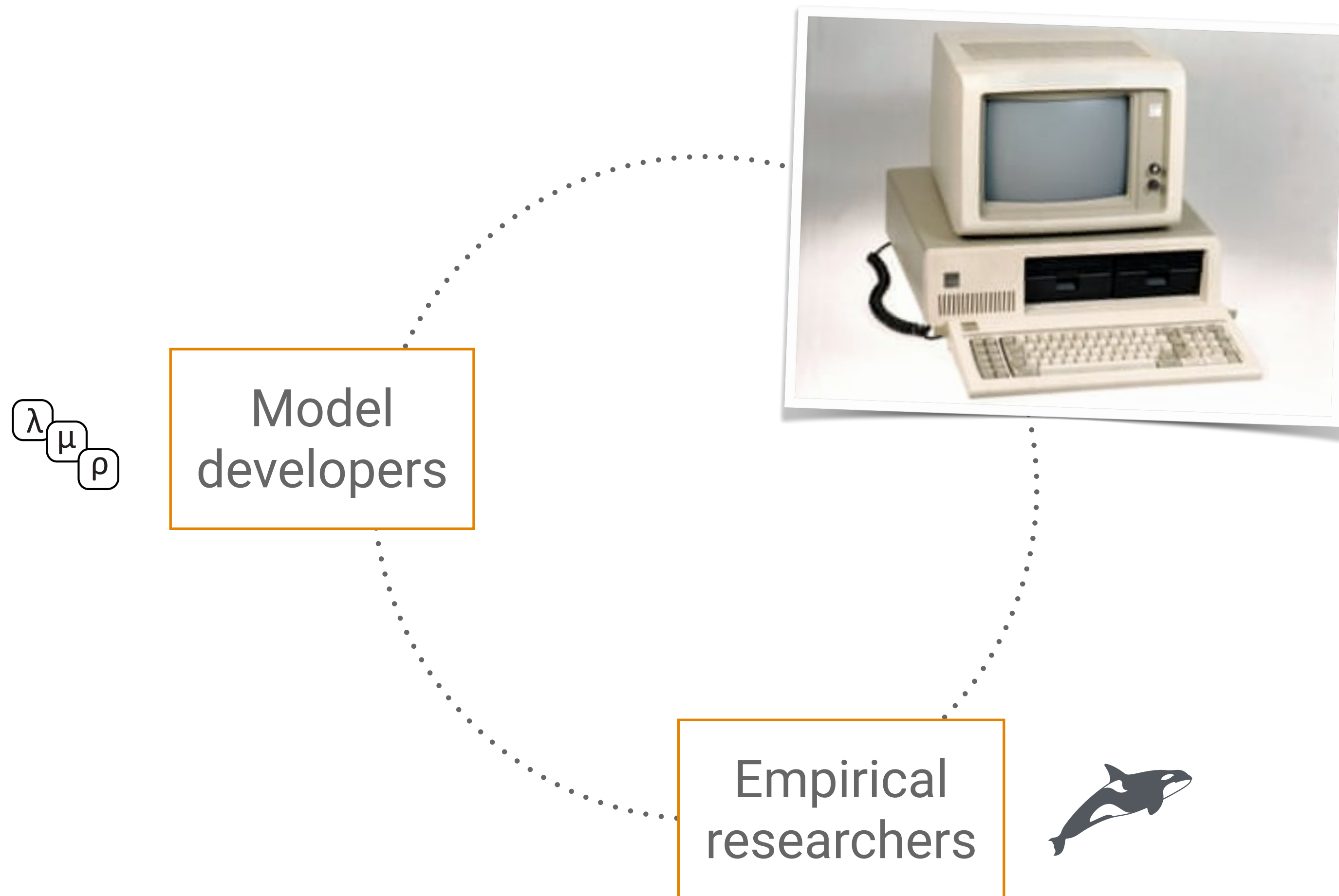
Intro to BEAST2

Phylogenetic inference — the old way

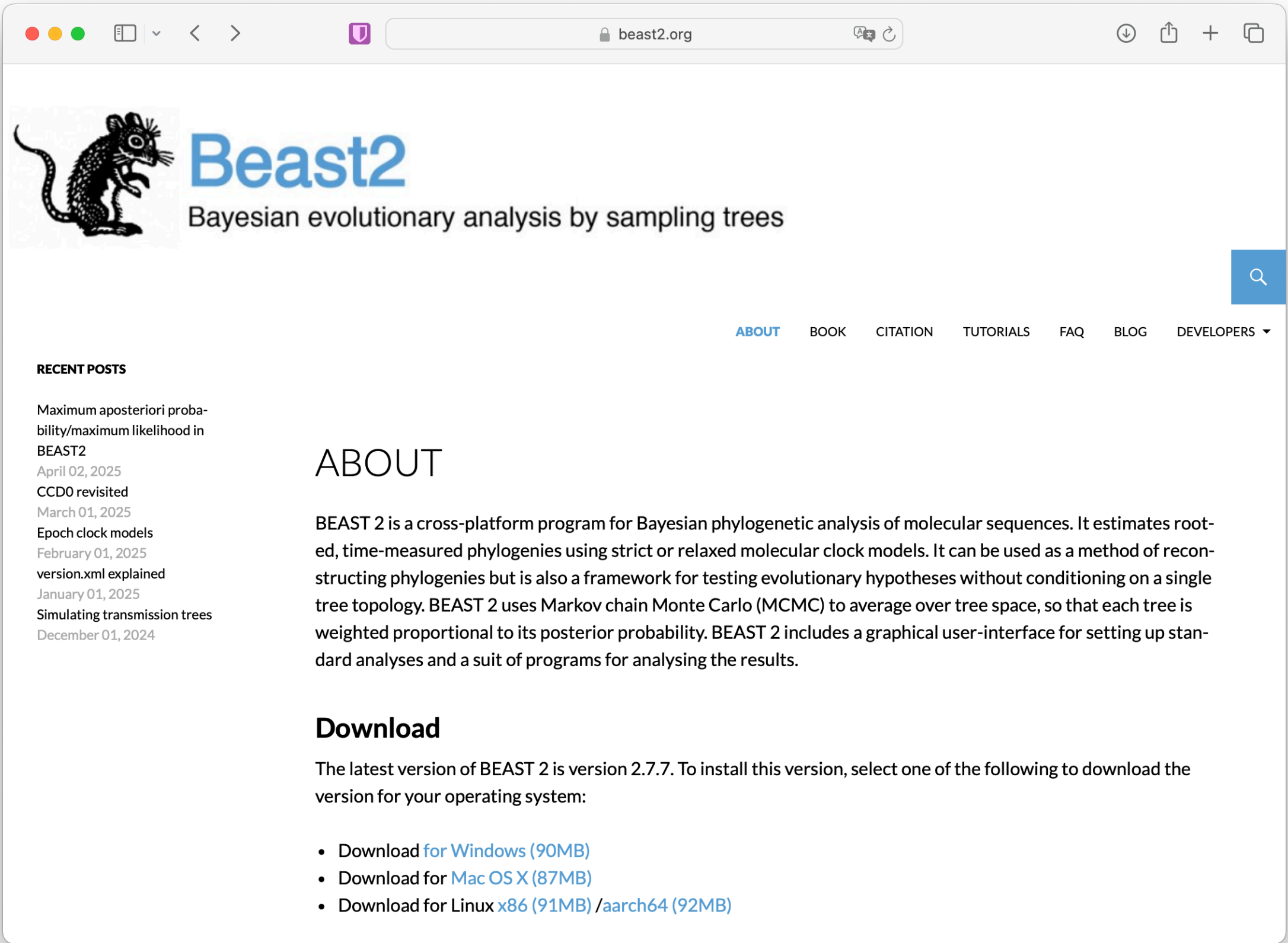


What we might call a
“**black box**” approach

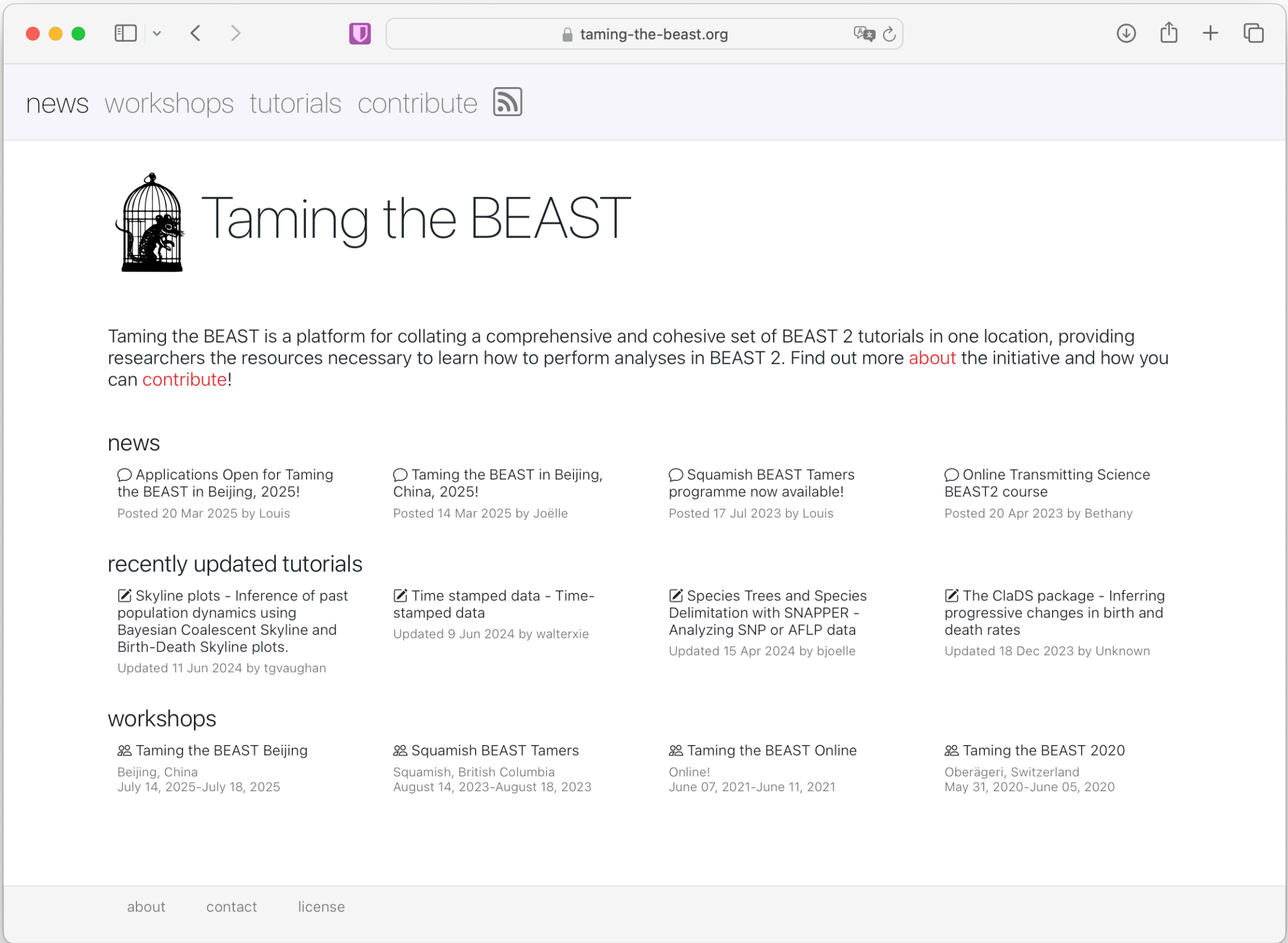
Phylogenetic inference — a better way?



The goal is to bring researchers with different expertise together, increase transparency, and do better research



www.beast2.org



taming-the-beast.org

BEAST2

Also designed with extendability and flexibility in mind

Also developed and supported by a large international team of developers

Has a suite of apps that can be used to generate input files and analyse the output

www.beast2.org

Beast2

Bayesian evolutionary analysis by sampling trees



[Scots poem](#) - also the [BEAST2](#) logo!

BEAST2 toolkit and work flow



Step 3a. Examine
you log files using
Tracer



Step 4. Examine your
summary tree in **FigTree**



Step 1. generate the
xml file in **BEAUi**



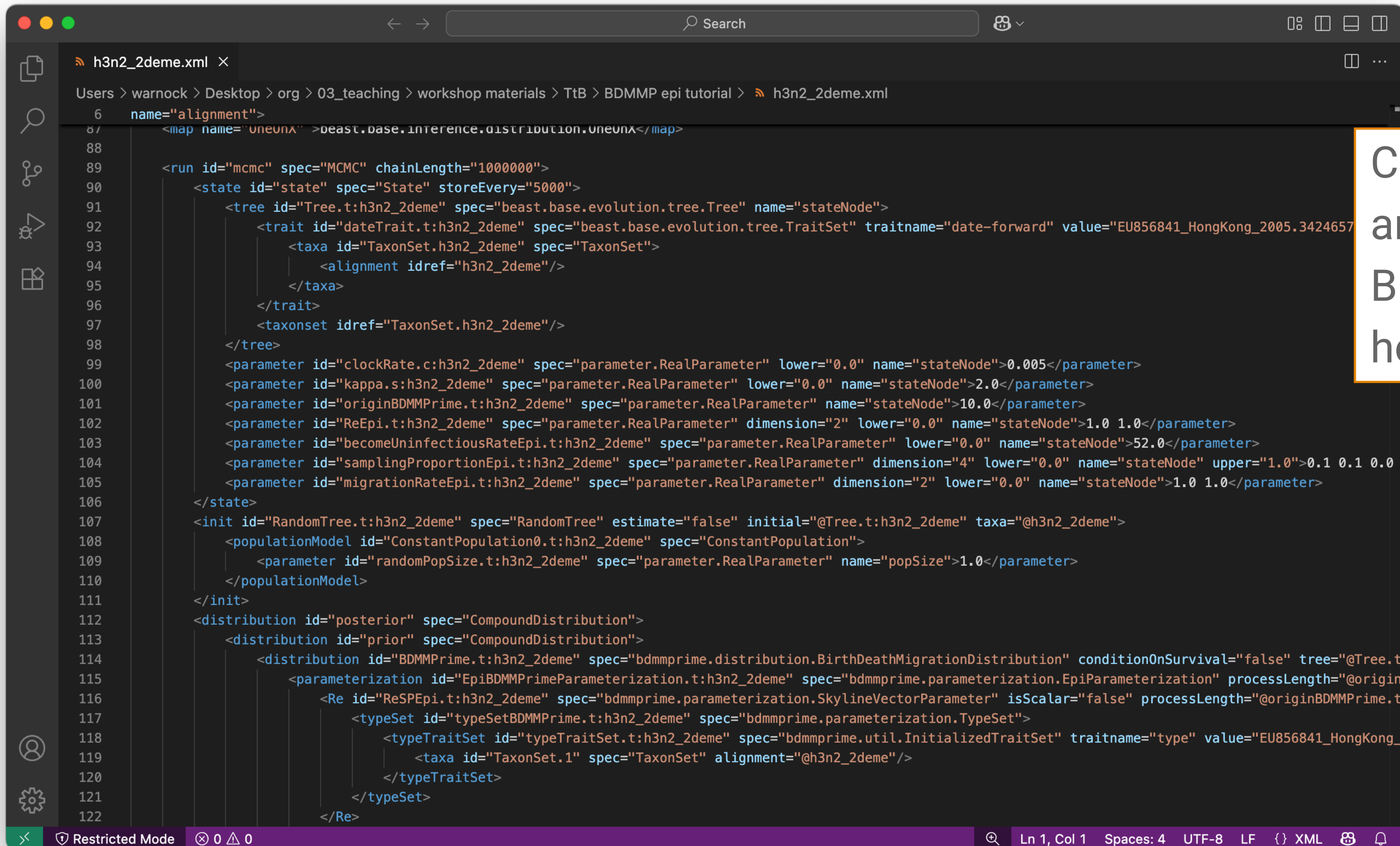
Step 2. run your
analysis in **BEAST**



Step 3b. Generate a
summary tree using
TreeAnnotator

Step... any other
downstream analysis

BEAST2 input: the XML file



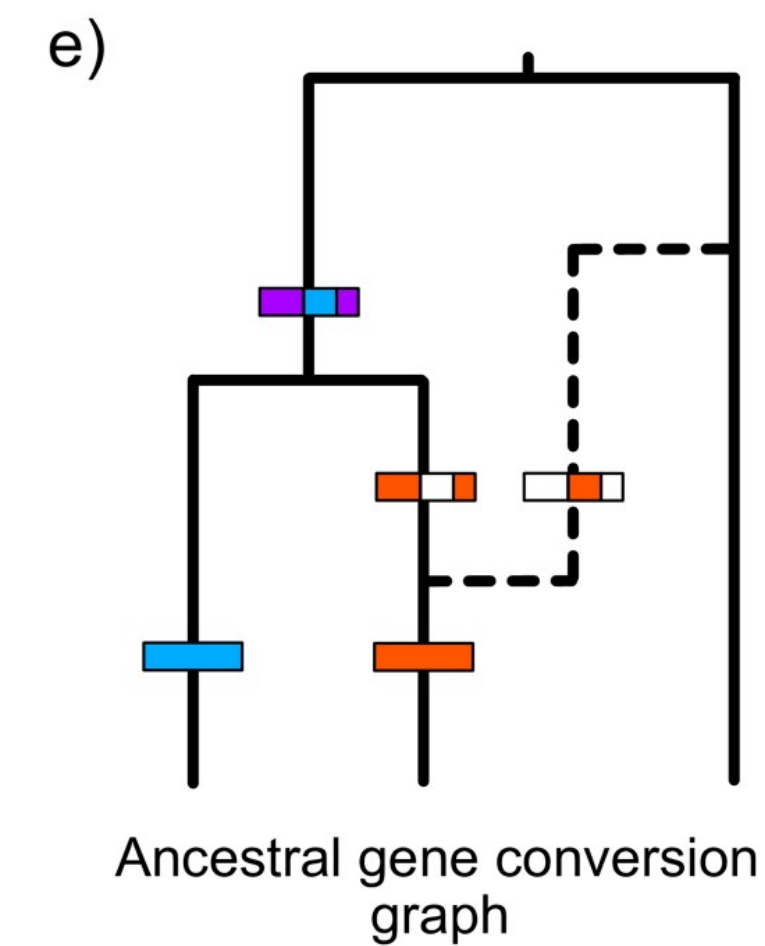
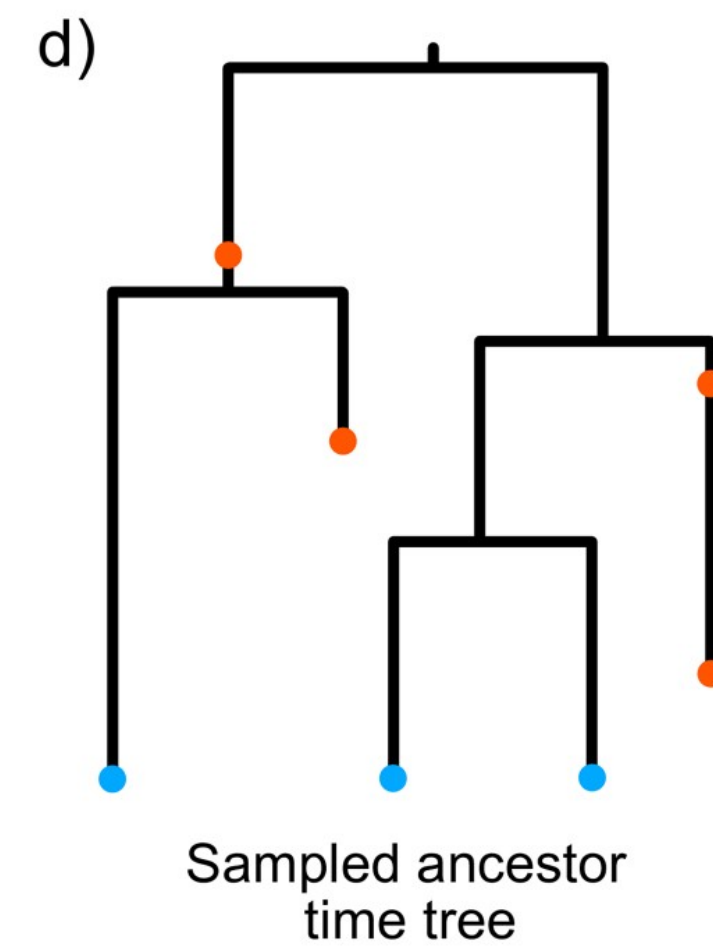
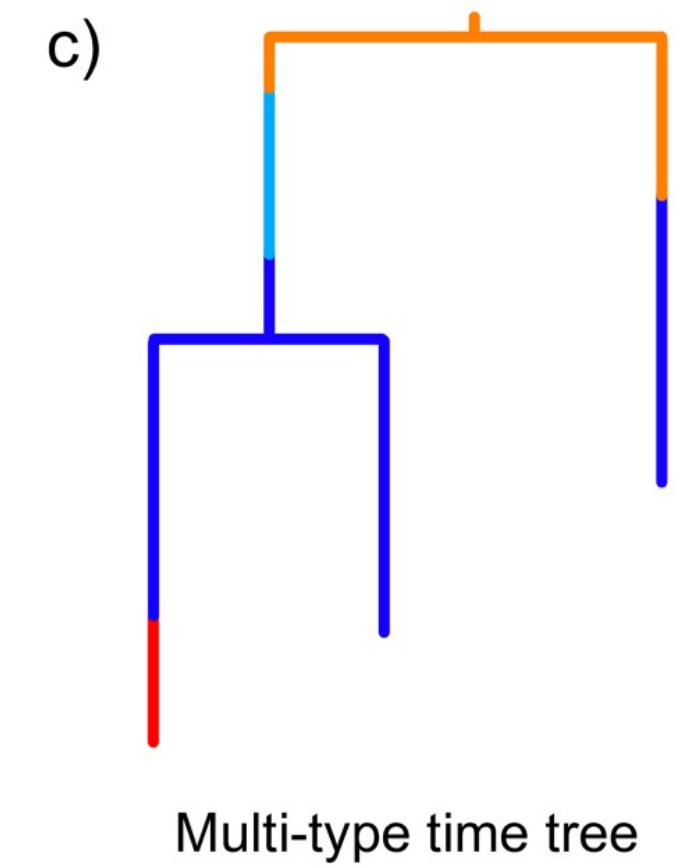
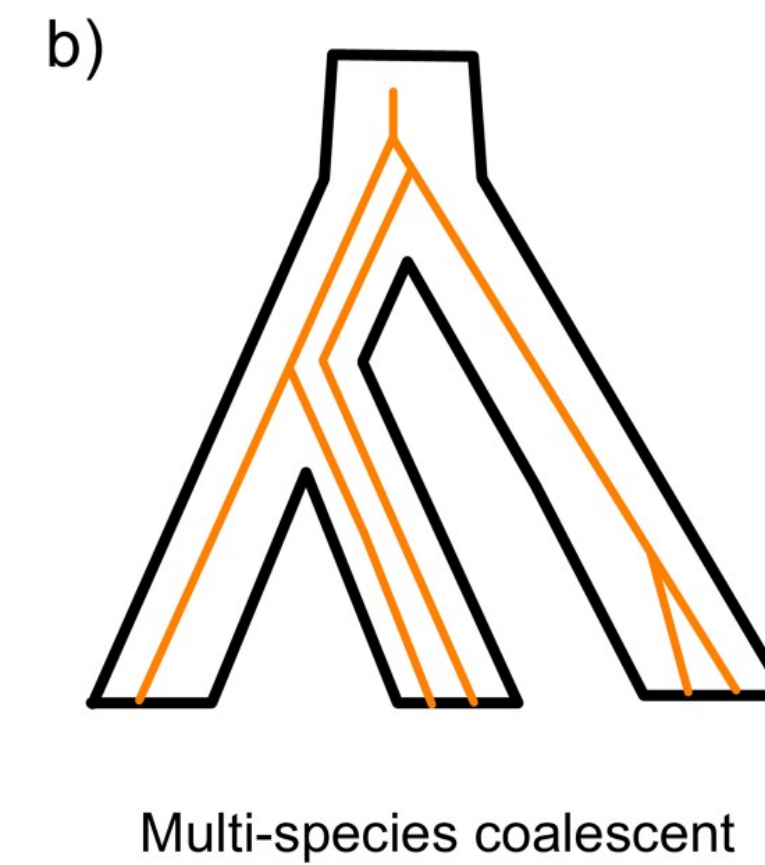
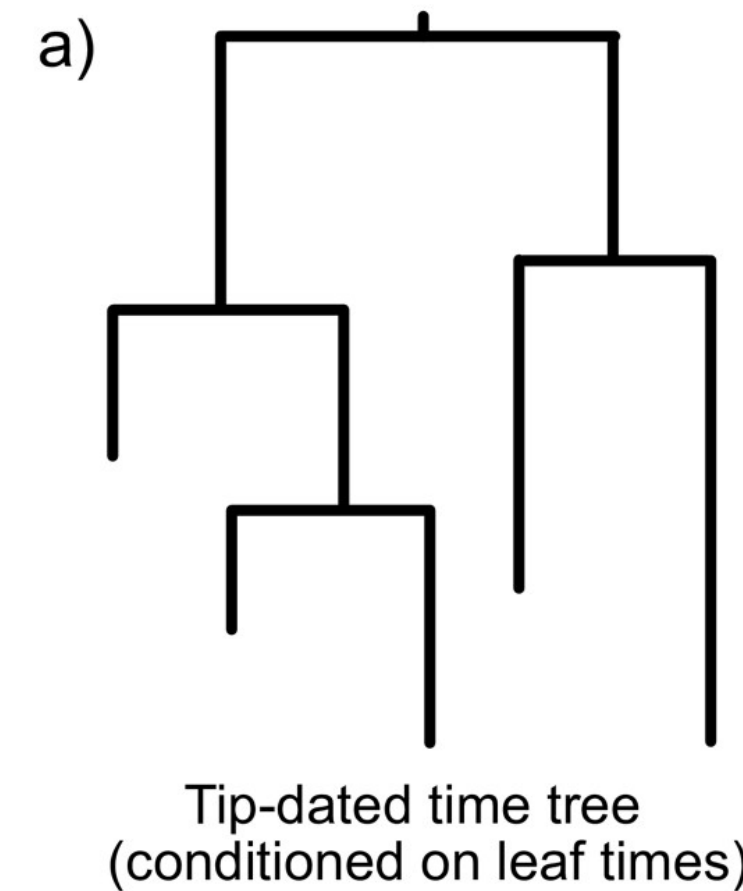
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87  <map name="oneunx" >beast.base.inference.distribution.oneunx</map>
88
89  <run id="mcmc" spec="MCMC" chainLength="1000000">
90    <state id="state" spec="State" storeEvery="5000">
91      <tree id="Tree.t:h3n2_2deme" spec="beast.base.evolution.tree.Tree" name="stateNode">
92        <trait id="dateTrait.t:h3n2_2deme" spec="beast.base.evolution.tree.TraitSet" traitname="date-forward" value="EU856841_HongKong_2005.3424657">
93          <taxa id="TaxonSet.h3n2_2deme" spec="TaxonSet">
94            <alignment idref="h3n2_2deme"/>
95          </taxa>
96        </trait>
97        <taxonset idref="TaxonSet.h3n2_2deme"/>
98      </tree>
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100      <parameter id="kappa.s:h3n2_2deme" spec="parameter.RealParameter" lower="0.0" name="stateNode">2.0</parameter>
101      <parameter id="originBDMMPPrime.t:h3n2_2deme" spec="parameter.RealParameter" name="stateNode">10.0</parameter>
102      <parameter id="ReEpi.t:h3n2_2deme" spec="parameter.RealParameter" dimension="2" lower="0.0" name="stateNode">1.0 1.0</parameter>
103      <parameter id="becomeUninfectiousRateEpi.t:h3n2_2deme" spec="parameter.RealParameter" lower="0.0" name="stateNode">52.0</parameter>
104      <parameter id="samplingProportionEpi.t:h3n2_2deme" spec="parameter.RealParameter" dimension="4" lower="0.0" name="stateNode" upper="1.0">0.1 0.1 0.0 0.0</parameter>
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117              <typeSet id="typeSetBDMMPPrime.t:h3n2_2deme" spec="bdmmpprime.parameterization.TypeSet">
118                <typeTraitSet id="typeTraitSet.t:h3n2_2deme" spec="bdmmpprime.util.InitializedTraitSet" traitname="type" value="EU856841_HongKong_2005.3424657">
119                  <taxa id="TaxonSet.1" spec="TaxonSet" alignment="@h3n2_2deme"/>
120                </typeTraitSet>
121              </typeSet>
122            </Re>

```

Caveat: if you want to use an option not available in BEAUti you have to learn how to edit the XML

A wide range of models and tree structures

Note all tree models in BEAST2 incorporate a temporal component



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